Mean rational complexity in integer bases: a uniform bound and a two-prime theorem

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Abstract

For a uniformly chosen residue modulo B^N, let its rational height be the smallest $\max(|a|, d)$ among congruent rational representations $a \equiv sd \pmod{B^N}$, with $d > 0$ coprime to the entire modulus. We prove a uniform upper bound strictly below $N/2+49/50$ for the mean base-B logarithm of this height. The additive correction can be removed for prime-power bases, and is strictly negative for bases with exactly two distinct prime factors. The latter result combines explicit estimates with finite integer certificates covering arbitrary positive valuations. The correction cannot be removed for all bases: $B = 30, N = 2$ is a certified counterexample. An explicit lower estimate also gives an eventually positive family with radical 510510. The self-contained argument places the optimal global additive correction strictly between zero and 49/50; a classical Jacobsthal bound further gives a correction strictly below 4/5. Its supremum reduces to finitely many bases, though neither its value nor finite-length attainment is determined here. Classical rational-reconstruction and lattice tools are distinguished from the new bounds. The companion specifies retained evidence, independent checks and full reconstruction commands.

1 Model, results and antecedents

Throughout, logarithms without a subscript are natural logarithms. All probability spaces in this paper contain *all* residues, not only invertible states.

Definition 1.1. For $M \geq 2$ and $s \in \mathbb{Z}/M\mathbb{Z}$, set

$$R_M(s) = \min_{\substack{d \in \mathbb{Z}_{>0},\ \gcd(d,M)=1 \\ a \in \mathbb{Z},\ a \equiv sd\ (M)}} \max(|a|, d).$$

For integers $B \geq 2, N \geq 1$, write $R_{B,N} = R_{B^N}$, with s uniformly distributed on $\mathbb{Z}/B^N\mathbb{Z}$, and put

$$E_{B,N} = \mathbb{E} \log_B R_{B,N}, \qquad \mu(M) = \mathbb{E} \log(R_M/\sqrt{M}).$$

Thus $E_{B,N} - N/2 = \mu(B^N)/\log B$.

The minimum exists: $d = 1$ with a centered representative of s gives $R_M(s) \leq \max(1, \lfloor M/2 \rfloor)$. In particular $R_M(0) = 1$. An admissible witness can be made primitive by dividing a, d by their gcd; the divisor is coprime to M. The full-unit condition on d is essential.

Theorem 1.2. *For every $N \geq 1$:*

1. *For every integer $B \geq 2$, $E_{B,N} < N/2 + 49/50$.*
2. *If B is a power of a prime, $E_{B,N} \leq N/2$.*
3. *If B has exactly two distinct prime factors, $E_{B,N} < N/2$.*

The binary case of the second assertion is the theorem of [1]; its proof is imported and attributed here. The odd-prime argument below is independent of limiting laws. The two-prime theorem includes $B = p^a r^b$ with arbitrary $a, b \geq 1$; it is not restricted to squarefree bases or equal valuations.

Theorem 1.3. *One has $E_{30,2} > 1$. If $\operatorname{rad}(B) = q_0 = 510510$ and $B^N \geq q_0^2$, then*

$$E_{B,N} > N/2 + \frac{1}{1000 \log B}.$$

Consequently the zero-correction bound does not hold for all bases.

Theorem 1.4. *Set*

$$c(B) = \sup_{N \geq 1}(E_{B,N} - N/2), \quad c_+(B) = \max(0, c(B)), \quad C^* = \sup_{B \geq 2} c(B).$$

Then $c_+(B) \to 0$ as $B \to \infty$, and $0 < C^* < 49/50$. There exists a finite B_0 such that $C^* = \max_{2 \leq B \leq B_0} c(B)$. This does not assert that the supremum over N is attained.

Question 3.2 of the survey [2, p. 20282] asks for an upper bound in general bases. Our answer concerns the explicitly defined rational height model. A connection integer in an FCSR realization may have the additional condition $d \equiv -1 \pmod{B}$, and register lengths may be rounded. We do not identify such quantities with this minimum for composite bases without a further equivalence theorem. The cited author draft of the monograph [3, Sections 19.3,21.3,21.5] treats coprime rational reconstruction and complexity estimates; its Section 19.1 uses a different connection-integer convention. Section locators here refer to that author draft, not a verified pagination of the published edition. The known lower estimates already give the leading coefficient $1/2$ for fixed bases. We do not claim that coefficient as a new result.

Chen–Winterhof [4] gives probabilistic binary results; its accessible preprint version has been checked, while the final published text has not been fully accessed for the source comparison reported here. The exact binary mean and variance results in [1] are prior results, not new claims here. Farey fractions with excluded denominators were studied by Boca–Heersink–Spiegelhalter [6]; congruence sections by Heersink [7]; smallest-denominator laws and logarithmic moments by Marklof [8, 9]. Those results are relevant antecedents, but an unmarked law cannot simply be rescaled to impose all the full-unit restrictions. Priority of the present bounds is not certified by the source comparison presented here.

2 A lattice upper bound

For a squarefree integer q, let $J(q)$ be the maximum gap between successive integers coprime to q, including the cyclic gap modulo q. Let λ_s be the shortest Euclidean length of a nonzero vector in

$$\Lambda_s = \{(a, d) \in \mathbb{Z}^2 : a \equiv sd \pmod{M}\}, \qquad \det \Lambda_s = M.$$

Lemma 2.1. *For $q = \operatorname{rad}(M)$,*

$$R_M(s) \leq \sqrt{1 + J(q)^2/3}\, \frac{M}{\lambda_s}.$$

Also $J(q) \leq 2 \cdot 4^{\omega(q)}$, uniformly over squarefree q.

Proof. A shortest vector u is primitive in the lattice and extends to a basis u, v. After subtracting an integer multiple of u, write $v = \theta u + v_\perp$, where $|\theta| \leq 1/2$ and $\|v_\perp\| = M/\lambda_s$. Minimality of u gives $\lambda_s^4 \leq 4M^2/3$. The second coordinates of u, v have gcd one, since the

lattice contains $(s, 1)$. If a prime dividing q divides the second coordinate of u, it cannot divide that of $v + ju$. For every other prime, precisely one residue class of j is forbidden. By CRT this is a translate of the unit pattern modulo a divisor of q. Its maximum gap is at most $J(q)$. Choose an allowed integer j within $J(q)/2$ of $-\theta$. Then $v + ju$ has a full-unit second coordinate and

$$\|v + ju\|^2 \le M^2/\lambda_s^2 + (J(q)^2/4)\lambda_s^2 \le (1 + J(q)^2/3)M^2/\lambda_s^2.$$

Changing the sign makes the denominator positive. If the denominator of u is already a unit, u itself is available; the same bound holds because $\lambda_s^2 \le 2M/\sqrt{3}$ and $J(q) \ge 1$.

For the second assertion, inclusion–exclusion shows that an interval of L consecutive integers contains at least $L\varphi(q)/q - 2^{\omega(q)}$ units. Since $\varphi(q)/q \ge 2^{-\omega(q)}$, $L = 2 \cdot 4^{\omega(q)}$ makes this quantity positive. Every such interval contains a unit. □

Lemma 2.2. *For $1 \le Y < M$, $\mathbb{P}(\lambda_s \le Y) \le 3Y^2/M$. Consequently, with $Z_M = R_M/\sqrt{M}$,*

$$\mathbb{P}(Z_M > z) \le \min(1, (J(q)^2 + 3)z^{-2}), \qquad \mu(M) \le \frac{\log(J(q)^2 + 3) + 1}{2}. \tag{2.1}$$

Proof. A horizontal nonzero lattice vector has length at least M. For $1 \le d \le Y$, put $g = \gcd(d, M)$. An integer a is incident with g residues s if $g \mid a$, and with none otherwise. Hence the number of incidences in the disk is at most

$$\sum_{1 \le d \le Y} (2\sqrt{Y^2 - d^2} + \gcd(d, M)) \le (1 + \pi/2)Y^2 < 3Y^2.$$

Here $\gcd(d, M) \le d$, $\sum_{d \le Y} d \le Y^2$ for $Y \ge 1$, and the decreasing semicircle sum is bounded by its integral. The union bound and Lemma 2.1 give the tail bound. When its threshold is below one there is no nonzero integral vector; when it is at least M, the stated probability bound is trivial. Finally

$$\mathbb{E}\log Z_M \le \mathbb{E}(\log Z_M)_+ \le \int_0^\infty \min(1, Ke^{-2t})\,dt = (\log K + 1)/2, \quad K = J(q)^2 + 3.$$

□

For two primes a slightly sharper tail is available:

$$\mathbb{P}(Z_M > z) \le \min(1, 6z^{-2}). \tag{2.2}$$

Indeed, if neither denominator of u, v is a full unit, the two bad-prime sets must be complementary. Then both $u + v$ and $u - v$ have unit denominators. The smaller Euclidean norm is at most $\sqrt{\|u\|^2 + \|v\|^2 - 2|\langle u, v\rangle|} \le \sqrt{2}M/\lambda_s$ by the reduced-basis bounds. If one basis vector is a unit the same inequality follows directly. Combining with the disk count proves (2.2).

A first uniform bound. Prime-power bases satisfy the stronger second assertion, proved below. Suppose q has at least two primes and $q \ge 10$. If q is odd, the units $1, (q-1)/2, (q+1)/2, q-1$ show $J(q) \le q/2$. If $q = 2d$ is even and $d \ge 5$ is odd, the units $1, d-2, d+2, 2d-1$ give the same conclusion. Thus

$$J(q)^2 + 3 \le 7q^2/25, \qquad e(J(q)^2 + 3) < q^2,$$

using $e < 3$. Equation (2.1) gives $\mu(B^N) < \log q \le \log B$. For $q = 6$, $J(6) = 4$; if $B \ge 12$, $19e < 57 < 144 \le B^2$. For $B = 6$, (2.2) gives $\mu(M) \le (\log 6 + 1)/2 < \log 6$. All constants are independent of N. □

Proposition 2.3 (An explicit common correction)**.** *For all integer bases $B \geq 2$,*

$$c(B) \leq \max(0, w, 39/40) < 49/50, \qquad w = 1 + \frac{\log(209/300)}{24 \log 2}.$$

In particular $C^* < 49/50$ and Theorem 1.2(1) holds.

Proof. The prime-power and two-prime arguments of Sections 4 and 5 do not use this proposition; they give $c(B) \leq 0$ when $k = \omega(q) \leq 2$, where $q = \mathrm{rad}(B)$. For the other bases, (2.1) gives

$$c(B) \leq \frac{\log(e(J(q)^2 + 3))}{2 \log q}.$$

If $3 \leq k \leq 5$, then $q \geq 30$ and $J(q) \leq \min(q/2, 2048)$. For $q \leq 4096$, using $e < 11/4$,

$$\frac{e(J(q)^2 + 3)}{q^2} < \frac{11}{4}\left(\frac{1}{4} + \frac{3}{30^2}\right) = \frac{209}{300}.$$

Put $\delta = -\log(209/300) > 0$. Since $2 \log q \leq 24 \log 2$, the common upper bound is $c(B) \leq w$. For $q \geq 4096$, the fixed numerator $\log(e(2048^2 + 3))$ and the increasing denominator give the same bound, by evaluating at $q = 4096$. Moreover

$$\delta > \frac{91}{300} + \frac{1}{2}\left(\frac{91}{300}\right)^2 = \frac{62881}{180000} > \frac{42}{125}, \qquad 24 \log 2 < \frac{84}{5}.$$

Thus $\delta/(24 \log 2) > 1/50$, and $w < 49/50$.

For $k \geq 6$, the product of the first six distinct primes is at least 30030, and every further prime is at least $17 > 16$. Hence $q \geq 30030\, 16^{k-6}$, while $J(q)^2 + 3 \leq 5\, 16^k$. The elementary enclosures $\log 30030 > 10$, $2/3 < \log 2 < 7/10$, $\log 5 < 17/10$ give

$$c(B) < \frac{(14/5)k + 27/10}{20 + (16/3)(k-6)} \leq \frac{39}{40} < \frac{49}{50}.$$

The rational function equals $39/40$ at $k = 6$ and decreases; its derivative numerator is -48. For the only additional constant, $e < 11/4$ and $11^{10} < 30030\, 4^{10}$ imply $\log 30030 > 10$. Taking the supremum over bases uses the two *common* caps w and $39/40$, rather than merely a pointwise strict inequality. No optimum is identified. □

3 Exact central counts and certified logarithms

Let $U_m(T) = \#\{1 \leq d \leq T : \gcd(d, m) = 1\}$. For $T^2 \leq M$ and a proper divisor $g \mid M$, put $m = M/g$ and

$$P_g(T) = \sum_{\substack{e \leq T/g \\ \gcd(e, M) = 1}} \mu_{\mathrm{Mob}}(e)\, U_m\left(\left\lfloor \frac{T}{ge} \right\rfloor\right) U_q\left(\left\lfloor \frac{T}{e} \right\rfloor\right), \quad q = \mathrm{rad}(M). \tag{3.1}$$

Here μ_{Mob} denotes the Möbius function. These count primitive positive witnesses $a = ga'$, with $\gcd(a', m) = 1$ and full-unit denominator. Within either sign no two such primitive fractions coincide modulo M: their determinant has absolute value at most $T^2 - 1 < M$. Opposite signs can collide only when $af + ed = M$; the possible value $2M$ at $T^2 = M$ would force all four components equal to T, which cannot give a unit denominator. Write $J_g(T)$ for these collisions in the gcd-g state layer. Then

$$C_g(T) = 2P_g(T) - J_g(T), \qquad C(T) = 1 + \sum_{g \mid M,\ g < M} C_g(T). \tag{3.2}$$

The one is the zero state and must not be absorbed into a nonzero layer.

For completeness the collisions are finite exact computations: enumerate ordered full-unit $d, f \le T$ with $\gcd(d, f) = 1$, set $a_0 = Mf^{-1} \bmod d$ with $0 \le a_0 < d$, and $e_0 = (M - fa_0)/d$. All solutions are $a = a_0 + kd,\ e = e_0 - kf$. Intersect the two closed integer ranges $1 \le a, e \le T$; assign surviving pairs to $\gcd(a, M)$, enforcing primitivity. This handles endpoints, signs and empty phases exactly.

3.1 Lower CDFs outside the central range

For $M = p^a r^b$, $p < r$, a nonzero state in layer g is $s = gt$, where t is a unit modulo $m = M/g$. For $m > 2$, choose the centered inverse magnitude h and its sign $\varepsilon \in \{1, -1\}$, so that $1 \le h \le H = \lfloor (m-1)/2 \rfloor$ and $th \equiv \varepsilon \pmod m$. The witness $(\varepsilon g, h)$ is admissible if h is a full unit. If m is a pure power of one prime u and the missing prime v divides h, replacing h by $m - h$ repairs the denominator and changes the numerator sign. Put

$$V(x) = \lfloor x/v \rfloor - \lfloor x/(pr) \rfloor .$$

For $T \ge g, m > 2$ the resulting lower CDF in that layer is

$$W_g(T) = 2U_q(\min(T, H)) + 2\mathbf{1}_{\{m \text{ pure}\}} \max(0, V(H) - V(\max(0, m - T - 1))). \tag{3.3}$$

For $T < g$ it is zero. When $m = 2$ the layer contains one state, with witness height g. There is also the direct-denominator-one bound

$$B_g(T) = 2U_m(\min(\lfloor T/g \rfloor, H)) \tag{3.4}$$

for $T \ge g, m > 2$, with the same exceptional conventions. Each layer has the deterministic height cap

$$K_g = \min(M/p,\ g\lfloor m/2 \rfloor,\ \max(g, D_m)), \tag{3.5}$$

where $D_m = 1$ for $m = 2$, $D_m = H$ when m has both primes, and $D_m = \max(H, m - v)$ when m is pure. The first cap follows from the circle lemma below; the others follow from the displayed witnesses. Bounds from the same layer are combined by a maximum, not a sum.

Lemma 3.1 (Circle cap)**.** *For M with smallest prime divisor p, $R_M(s) \le \max(M/p, p - 1)$. If $M/p \ge p - 1$, the bound is M/p, and equality occurs for $s = M/p$. This condition holds for every two-prime modulus and every prime power of exponent at least two.*

Proof. Place $0, s, \ldots, (p-1)s$ on the circle of circumference M. If two coincide their index difference supplies numerator zero and a denominator between one and $p - 1$. Otherwise their smallest cyclic gap is at most M/p, again with index difference between one and $p - 1$. That denominator is coprime to M, proving the maximum bound. For $s = M/p$, every unit denominator is nonzero modulo p; its centered product with s has absolute value at least M/p. □

For $T^2 > 2M$, delete nonunit denominators from the Farey sequence of order T. Adjacent deleted denominators with complementary bad primes have unit difference; taking that difference repairs one neighbor. A gap can remain uncovered only next to a bad center $d < M/T$. Its two retained neighbors have denominators $> T - jd$, where $j = 1$ if $pr \mid d$, and $j = 2$ otherwise. These are positive because $T^2 > 2M$. Its open length in residue coordinates is at most $2(M - Td)/(d(T - jd))$. One extra integer per open hole gives

$$S(T) \le \sum_{\substack{d \le \lfloor (M-1)/T \rfloor \\ \gcd(d,M) > 1}} \varphi(d) + \sum_{\substack{d \le \lfloor (M-1)/T \rfloor \\ \gcd(d,M) > 1}} \frac{2\varphi(d)(M - Td)}{d(T - jd)}. \tag{3.6}$$

This is an upper survival bound, not an equality. The strict start $\lfloor\sqrt{2M}\rfloor + 1$ avoids the removable equality endpoint. The same Farey repair and length argument underlies the analytic estimates in the appendix.

3.2 Pointwise histograms for the initial finite bridges

For a two-prime modulus M, the base-six diagonal, sixth-power and squarefree-prefix batches use a pointwise construction, distinct from the aggregate CDF calendar. Put

$$k = \lfloor\sqrt{M}\rfloor, \qquad \tau = \lfloor\sqrt{2M}\rfloor + 1, \qquad D_1(s) = \max(1, \min(s, M - s)).$$

The closed central enumeration gives the exact height of every state with $R_M(s) \le k$, including zero separately. Apply the proved far-cover construction at τ, retaining its actual full-unit neighbors and phases, and let $\mathcal{H}$ be the union of the open exceptional holes. Outside $\mathcal{H}$, set $U(s) = \min(\tau, D_1(s))$, since the cover proves $R_M(s) \le \tau$. Inside $\mathcal{H}$, start with $D_1(s)$ and lower $U(s)$ only by heights $\max(|a|, d)$ of explicitly checked pairs

$$d > 0, \qquad \gcd(d, M) = 1, \qquad a \equiv sd \pmod{M}.$$

Every such pair is an admissible upper witness; overlaps are handled by minima. Restore the exact central heights wherever available. Consequently $R_M(s) \le U(s)$ pointwise, and its histogram has mass M.

For example, at a reduced exceptional center c/v, put $\Delta = Mc - vs$ and $\varepsilon = \operatorname{sign}(\Delta)$, with $\varepsilon = 1$ if $\Delta = 0$. For any positive full-unit d with $cd \equiv \varepsilon \pmod{v}$, the integer

$$a = \varepsilon(M - |\Delta|d)/v$$

satisfies $a - sd = M(\varepsilon - cd)/v$. Thus it is a legal witness; its optimality is neither assumed nor needed. The implementation checks integrality, congruence and denominator marks.

For $M = pr$, $p < r$, cap U by r, using Lemma 3.1. When $r \le \tau$, that cap suffices off the exact central set. For the actual histograms in these batches, the separately checked integer sign premise $Q < 0$, together with $R_M \le U$ and the outward logarithm enclosures below, proves the required negative centered mean. Pointwise domination alone does not imply that sign. For the diagonal moduli $M = 6^n$, $2M$ is nonsquare, so the historical threshold $\lceil\sqrt{2M}\rceil$ equals the strict threshold τ.

3.3 Integer logarithmic certificates

A calendar contains all integers up to an initial k, then steps by $T \mapsto \lceil 257T/256\rceil$, with every layer cap, the far-tail start, and the terminal M/p inserted. At each calendar point combine (3.2) when applicable, the layer witness bounds, and (3.6); take cumulative maxima. Filling the gaps by the previous CDF value produces a lower CDF $\underline{C} \le C$. Its increments n_h give a dominating height histogram:

$$\sum_h n_h = M, \qquad \mathbb{E}\log R_M \le M^{-1}\sum_h n_h \log h.$$

The increments are not claimed to be the true height distribution.

Use outward integer ticks $L_S(x) \le S\log x \le U_S(x)$ and retain

$$Q = 2\sum_h n_h U_S(h) - M L_S(M). \tag{3.7}$$

Then $Q < 0$ proves $\mu(M) < 0$, including odd M. The identity

$$\log x = 2\sum_{j=0}^{K-1} \frac{z^{2j+1}}{2j+1} + \varepsilon, \quad z = (x-1)/(x+1), \quad 0 \le \varepsilon \le \frac{2z^{2K+1}}{(2K+1)(1-z^2)}$$

for rational $x \ge 1$, with range reduction by powers of two, supplies exact rational enclosures. No floating logarithm or assertion-sensitive test is used for the sign. The independent checker uses a different series depth and tick scale. It independently reconstructs the initial boundary, terminal boundary and relevant boundary gcd strata. It consumes the intervening central arrays; the central lemma and producer are therefore essential parts of the proof. It is not a complete independent recount of every intermediate CDF value.

4 Prime powers

For odd $M = p^n$ let $a = \sqrt{M}$, $K = \left\lfloor a/\sqrt{2} \right\rfloor$. The central count is

$$C(T) = 1 + 2P_p(T), \qquad P_p(T) = \#\{u, d \le T : \gcd(u,d) = 1,\ p \nmid d\}, \quad T \le K.$$

Indeed $2T^2 < M$, so opposite primitive witnesses do not collide. Before primitivity there are at least $(1-1/p)T^2$ pairs. The divisor union bound gives

$$P_p(T) \ge (1-y)\{2 - (1-y^2)\zeta(2)\}T^2 - (1-y)T\log T, \qquad y = 1/p.$$

To see the error explicitly, a common divisor $d \ge 2$ must satisfy $p \nmid d$, and $\lfloor x \rfloor (\lfloor x \rfloor - \lfloor x/p \rfloor) \le (1-y)(x^2+x)$. Sum over d, use $\sum_{d \ge 2, p \nmid d} d^{-2} = (1-y^2)\zeta(2) - 1$ and $\sum_{d=2}^{T} d^{-1} \le \log T$. The elementary enclosure $\zeta(2) < 5/3$ follows by summing five terms and bounding the rest by an integral. Consequently

$$C(T) \ge cT^2 - dT\log T, \quad (c,d) = \begin{cases} (56/81, 4/3), & p = 3, \\ (5/8, 2), & p \ge 5. \end{cases} \tag{4.1}$$

The exact compensation identity is

$$\mu(M) = -D + P_+, \quad D = M^{-1}\int_1^a C(\lfloor x \rfloor)\frac{dx}{x}, \quad P_+ = M^{-1}\int_a^\infty S(\lfloor x \rfloor)\frac{dx}{x}. \tag{4.2}$$

Retaining the plateau $C(K)$ up to a yields

$$D \ge c(K^2/a^2)(1/2 + \log(a/K)) - 2cK/a^2 - d(K\log K/a^2)(1 + \log(a/K)). \tag{4.3}$$

For $a \ge 140$, $K/a \ge 11/16$, $K/a < 3/4$, $K^2/a^2 \ge 1/2 - \sqrt{2}/a$, and $\log 2/2 \le \log(a/K) \le 5/11$. Using $2/3 \le \log 2 \le 1$ and $\sqrt{2} < 3/2$ gives

$$D \ge \begin{cases} 70/243 - [56/27 + (16/11)\log a]/a, & p = 3, \\ 25/96 - [15/8 + (24/11)\log a]/a, & p \ge 5. \end{cases} \tag{4.4}$$

For integer $T \ge a$, the holes next to reduced Farey centers with $p \mid d, d < M/T$ have retained neighbors $b_L, b_R > T - d$. Their open lengths are at most $2(M - dT)/(d(T-d))$. Let $\kappa = (p-1)/p^2$, $t = T/a$, and

$$F(t) = \sum_{j \ge 1} \frac{t^{-2j}}{j(j+1)} = 1 + (t^2-1)\log(1 - t^{-2}), \quad F(1) = 1.$$

Summation of lengths and one integer per hole, using $\varphi(pk) \le (p-1)k$, gives

$$S(T)/M \le 2\kappa F(t) + \frac{\kappa a^2}{2T^2} + \frac{1-1/p}{2T}. \tag{4.5}$$

One can check the rounding term directly: the hole count is bounded by $(p-1)L(L+1)/2$, $L = \lfloor M/(pT) \rfloor$. The decreasing length summand is integrated with spacing p. Since F decreases and $\int_1^\infty F(t)\,dt/t = (\zeta(2)-1)/2$, integrating the initial interval $[a, \lceil a \rceil]$ and retaining the first summand gives

$$P_+ \le \begin{cases} 11/54 + 17/(9a) + 1/(3a^2), & p = 3, \\ 11/75 + 19/(10a) + 1/(2a^2), & p \ge 5. \end{cases}$$

Thus

$$\mu(M) \le \begin{cases} -41/486 + [107/27 + (16/11)\log a]/a + 1/(3a^2), & p = 3, \\ -91/800 + [151/40 + (24/11)\log a]/a + 1/(2a^2), & p \ge 5. \end{cases} \tag{4.6}$$

At $a = 140$, using $\log 140 < 5$, both upper bounds are negative: the positive margins are at least $214289/52390800$ and $3811/431200$, respectively. All error terms decrease thereafter.

For $M = p \ge 5$, choose $k = \lfloor \sqrt{p} \rfloor$. Among the $(k+1)^2 > p$ pairs $0 \le u, v \le k$, two have the same $u - sv$; their difference has nonzero denominator at most $k < p$. Thus $R_p \le k < \sqrt{p}$. For $M = p^2$, the circle argument gives $R_M \le p = \sqrt{M}$. Only the 21 entries in Table 1 remain below 19600. Here $H = \sum_s R_M(s)$ and the exact inequality $H^2 < M^3$ implies $\mu(M) < 0$ by Jensen. The companion computes these minima without an assumed mean bound.

Prime	Exponents n	Sums H, in order
3	1–8	3,17,105,577,3107,16527,86977,455939
5	3–6	1009,11729,134011,1513793
7	3–5	4447,84975,1595511
11	3–4	32803,1220493
13	3	68697
17	3	225957
19	3	370511
23	3	866951

Table 1: The finite odd-prime bridge.

This proves Theorem 1.2(2) for odd prime powers. If $B = p^b$, replace n by bN; no new finite cases occur. For $p = 2$, use the exact theorem of [1].

5 The two-prime theorem

The appendix gives the full parametric compensation inequality and its refinements. Its proof uses the exact central formula, explicit errors for the unit-counting function, and the Farey survival estimate (3.6). It is an inequality at each finite M; no limiting approximation is substituted for a finite bound. In particular, for $M = p^a r^b$, it proves negativity above:

prime factors	sufficient modulus
$2, 3$	6^9
$2, r,\ r \ge 5$	$9\,000\,000$
$3 \le p < r$	2^{22}

(5.1)

The sharper uniform cutoffs, apart from the four radicals $6, 10, 14, 15$, are

$$M \ge C(p), \quad C(2) = 384^2, \quad C(3) = 512^2, \quad C(p) = 600^2 \ (p \ge 5). \tag{5.2}$$

Appendix E specifies all finite domains and reductions. The certificates are organized by domains of moduli, not assumed experimental monotonicity of means.

Proof of Theorem 1.2(3). First the analytic bounds and the finite bridges settle the four exceptional radicals $6, 10, 14, 15$ at every pair of positive valuations. The squarefree two-prime moduli are settled by a single-mark reduction, with 4596 smaller moduli certified explicitly. For all bases, the analytic cutoffs leave three sixth powers, three fourth powers, 35 cubes and 99 squares, as detailed in the appendix. These imply the assertion for every length $N \geq 2$. The stronger length cutoffs are reindexings of the same modulus, not new hypotheses on states.

For $N = 1$, write $M = p^a r^b$. If $\gcd(a, b) \geq 2$, it is already a higher power of a two-prime base. Remove squarefree moduli, the four settled radicals, and the moduli above (5.2). The resulting exact inventory has 19239 profiles. A larger-prime-exponent-one lemma pays 7857 profiles. The five explicitly proved analytic bounds pay another 6972. The remaining 4410 profiles split into the three disjoint certificate domains of sizes 1087,1422 and1901. Each retained integer Q in (3.7) is strictly negative. All inventories, prime tests, powers, exclusions and sign comparisons are finite integer operations. The domain identities and mass totals are checked in the companion. There is no unexamined valuation class. □

6 A lower estimate and positive corrections

For squarefree q, put

$$D_q = 2^{\omega(q)}, \quad \rho_q = \prod_{p|q} \frac{p}{p+1}, \quad \alpha_q = \frac{2\rho_q}{\zeta(2)}, \quad \beta_q = -\frac{1 + \log \alpha_q}{2}.$$

Lemma 6.1. *With $q = \mathrm{rad}(M)$ and $L = \sqrt{M/\alpha_q}$,*

$$\mu(M) \geq \beta_q - \left(\frac{2(D_q + 3)}{\sqrt{\alpha_q M}} + \frac{1}{M} \right) \log L. \tag{6.1}$$

Proof. Let $P_q(T)$ count positive primitive pairs $a, d \leq T$ with $\gcd(d, q) = 1$. Möbius inversion gives

$$P_q(T) = \sum_{\substack{e \leq T \\ \gcd(e,q)=1}} \mu_{\mathrm{Mob}}(e)\, \lfloor T/e \rfloor\, U_q(\lfloor T/e \rfloor).$$

Inclusion–exclusion gives $|U_q(x) - (\varphi(q)/q)x| \leq D_q$. Expanding each product, summing $e^{-1} \leq 1 + \log T$, and bounding the tail of $\sum e^{-2}$ by $1/T$, yields

$$\left| P_q(T) - \frac{\rho_q}{\zeta(2)} T^2 \right| \leq (D_q + 3) T (1 + \log T). \tag{6.2}$$

The main coefficient follows from $(\varphi(q)/q) \sum_{\gcd(e,q)=1} \mu_{\mathrm{Mob}}(e) e^{-2} = \rho_q/\zeta(2)$. These bounds also cover $T = 1$. Every admissible fraction can be reduced to a primitive one, so $C(T) \leq 1 + 2P_q(T)$ by the union bound; injectivity is not needed. Use the resulting upper bound for $C(\lfloor x \rfloor)$ on $1 \leq x \leq L$ in $\mathbb{E} \log R_M = \int_1^\infty (1 - C(\lfloor x \rfloor)/M)\, dx/x$, and discard the nonnegative remainder. Integrating gives β_q as the main centered term. The error integral is at most $2(D_q + 3) L \log L/M + \log L/M$. The omitted lower endpoint main term is nonnegative, giving (6.1). □

For fixed B, this bound and (2.1) yield $E_{B,N}/N \to 1/2$, consistently with the antecedent in [3].

6.1 The exact finite obstruction

The direct height histogram for $M = 900$ has total mass 900 and height sum 34007. If $P = \prod_{s=0}^{899} R_{900}(s)$, integer multiplication gives

$$2 \cdot 30^{900} \le P < 3 \cdot 30^{900}. \tag{6.3}$$

The second bound is recorded for reproducibility, not needed for the sign. Hence $\mathbb{E} \log_{30} R_{900} = \log_{30} P/900 > 1$. An independent algorithm enumerates admissible denominators and centered numerators directly. Replacing full-unit denominators by denominators coprime to only one prime changes many minima; that wrong model is not used. The decimal $1.0002564533304775\ldots$ is descriptive; the proof is the integer comparison (6.3). It refutes a possible strengthening, not a conjecture stated in Question 3.2.

6.2 An eventually positive family

For $q_0 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 = 510510$,

$$\rho_{q_0} = 12155/41472.$$

For $M \ge q_0^2$, the main lower term in (6.1) satisfies $\beta_{q_0} > 7/500$, and its error is $< 13/1000$. We give rational checks to make this explicit. The elementary enclosures $e < 87/32$ and $\zeta(2) > 41/25$ imply

$$e\alpha_{q_0} < 1 - \frac{256169}{9068544}, \qquad \beta_{q_0} > \frac{256169}{18137088} > \frac{7}{500}.$$

The first follows from a factorial-series upper bound for e, the second from a partial sum for $\zeta(2)$. Also $\zeta(2) < 5/3$ gives $\alpha_{q_0} > 2431/6912 > 7/20 > (59/100)^2$. At $M = q_0^2$, $\log L < 14$ and $q_0 > 510000$, so

$$\left(\frac{2(128+3)}{\sqrt{\alpha_{q_0} M}} + \frac{1}{M}\right) \log L < \left(\frac{262}{(59/100)510000} + \frac{1}{510000^2}\right) 14 < \frac{13}{1000}.$$

The functions $\log(\sqrt{M/\alpha})/\sqrt{M}$ and $\log(\sqrt{M/\alpha})/M$ decrease on this range. Thus $\mu(M) > 1/1000$, proving the family in Theorem 1.3. No minimal radical or minimal cutoff is claimed. For radicals containing q_0, an eventually positive conclusion also follows from (6.1) with that radical's own constants.

7 The global additive optimum

Proof of Theorem 1.4. Let $k = \omega(q)$. Since the i-th smallest distinct prime is at least $i+1$, $q \ge (k+1)!$. Lemma 2.1 therefore gives $J(q) = q^{o(1)}$ uniformly over squarefree radicals: for $k \to \infty$,

$$\frac{\log J(q)}{\log q} \le \frac{\log 2 + k \log 4}{\log((k+1)!)} \longrightarrow 0;$$

for bounded k and $q \to \infty$, the numerator is bounded. Equation (2.1) gives

$$c_+(B) \le \frac{\log(J(\operatorname{rad} B)^2 + 3) + 1}{2 \log B}.$$

If the radical tends to infinity this goes to zero, since $\log q \le \log B$. If radicals stay bounded their numerator is bounded and $\log B \to \infty$. This proves $c_+(B) \to 0$.

The witness $c(30) \geq E_{30,2} - 1 > 0$ now excludes all sufficiently large bases from the global supremum. Hence only finitely many bases remain. Proposition 2.3 supplies the common cap $C^* \leq \max(0, w, 39/40) < 49/50$. Together with the positive witness, this gives $0 < C^* < 49/50$. A finite maximum of the quantities $c(B)$ is still a maximum of suprema over infinitely many lengths; finite-length attainment is not part of the conclusion. □

This elementary argument requires no hypothesis about extremal Jacobsthal gaps or prime distributions. It is not a practical algorithm for locating the maximizing base.

Corollary 7.1. *If $c(B) = C^*$, then B has at least three distinct prime factors and is not a perfect power. Equivalently, the gcd of its positive prime valuations is one.*

Proof. The classes with one or two primes have $c(B) \leq 0 < C^*$. For $B = A^e$, $e \geq 2$, equality of modulus and witness spaces gives

$$E_{A^e,N} - N/2 = \frac{1}{e}(E_{A,eN} - eN/2), \qquad c(A^e) \leq c(A)/e \leq C^*/e < C^*.$$

The supremum here is over a subset of the lengths for A; no maximizing finite length is asserted. Non-perfect-power does not mean squarefree. □

Corollary 7.2 (A classical Jacobsthal sharpening)**.** *Using Kanold's bound $J(q) \leq 2^{\omega(q)}$, as reported by Costello–Watts [5, Introduction, pp. 1–2], one has*

$$0 < C^* \leq \frac{\log 67 + 1}{2 \log 30} < \frac{5037}{6400} < \frac{4}{5}.$$

In particular, $E_{B,N} < N/2 + 4/5$ for all $B \geq 2, N \geq 1$.

Proof. The maximum cyclic gap $J(q)$ equals the least length of a consecutive integer interval guaranteed to contain a q-unit: a gap of length G has $G - 1$ intervening nonunits. Thus the cited bound uses precisely our convention, and applies to every radical, not only primorials.

Put $q = \mathrm{rad}(B)$ and $k = \omega(q)$. For $k \leq 2$, Theorem 1.2(2)–(3) gives $c(B) \leq 0$. For $k \geq 3$, equation (2.1), uniformly in N, gives

$$c(B) \leq \frac{\log(4^k + 3) + 1}{2 \log q}.$$

The first three distinct prime divisors have product at least 30, and each further prime is at least 7, hence at least 4. Consequently $q \geq 30\, 4^{k-3}$. Also $\log(4^k + 3) \leq k \log 4 + \log(67/64)$, so

$$c(B) \leq F(k) := \frac{k \log 4 + 1 + \log(67/64)}{2 \log 30 + 2(k-3) \log 4}.$$

The denominator is positive for real $k \geq 3$; the numerator of its derivative is $2 \log 4(\log 30 - \log 67 - 1) < 0$. Thus $F(k) \leq F(3) = (\log 67 + 1)/(2 \log 30)$. This is a common cap before either supremum over N or B.

For a rational enclosure, Appendix C gives $\log 2 < 7/10$ and $e < 11/4$. Together with $\log(1 + x) < x$,

$$\log 67 + 1 = 6 \log 2 + \log(67/64) + 1 < \frac{21}{5} + \frac{3}{64} + 1 = \frac{1679}{320}.$$

Since $11^{10} = 25937424601 < 28311552000 = 27000\, 4^{10}$, one has $e^{10} < 30^3$, hence $\log 30 > 10/3$. Therefore

$$\frac{\log 67 + 1}{2 \log 30} < \frac{1679/320}{20/3} = \frac{5037}{6400} < \frac{5120}{6400} = \frac{4}{5}.$$

Positivity follows from Theorem 1.3. □

This sharpening imports the cited classical gap bound; the self-contained 49/50 argument above remains valid independently. The argument does not determine the optimal correction or its extremizers.

8 Reproducibility and limitations

The repository companion contains the original finite mathematical premises, portable reconstruction programs, exact logarithmic sums, complete domain lists and a provenance map. The original default checks remain a short retained-evidence check, a fixed defining oracle and a four-case squarefree reconstruction. They are distinct from the complete reconstruction route provided by `scripts/reconstruct_all.py`. The reproduction guide specifies the commands, domains and limits of both routes.

The complete source-replay chain passed all 37 stages in each of normal and optimized Python modes. The squarefree prefix, which originally retained only digests and signs, now has all 4596 reconstructed upper histograms and their consumed logarithmic ledgers. These histograms cover 41585055 original envelope entries. A separate K32 logarithm implementation checked all 705257 consumed tick entries, involving 6929 distinct integers, and reconstructed every negative outward sum and its exact normalization. The retained full artifacts are indexed by `companion/reconstruction/index.json`; `scripts/check_complete_evidence.py` checks their declared complete coverage, byte identities and the squarefree logarithmic certificates.

An additional distinct central checker reconstructs all integer thresholds $1 \le T \le k = \lfloor \sqrt{M} \rfloor$ and every proper original gcd stratum for all nine aggregate batches, including their consumed validation rows. For each full-unit denominator $1 \le d \le k$, it enumerates every $1 \le a \le k$ and both signed residues $\pm ad^{-1} \pmod{M}$, retaining the smallest $\max(a, d)$ at each state. There is no $\gcd(a, d) = 1$ filter: nonprimitive pairs are legal witnesses too. Zero is one separate atom; coincident signed witnesses update the same state. The resulting central height bins give every stratum prefix. All source-produced central arrays are compared at every threshold, including explicit zero counts for omitted strata above k. The original endpoint-only checker descriptions elsewhere in the paper refer to the earlier retained-data checks, rather than this additional complete central comparison.

The same direct central enumeration also checks every central histogram prefix for all 4596 squarefree cases, the eight base-six cases and the three sixth-power cases. All nine aggregate and all three pointwise-domain checks completed in both normal and optimized modes. The pointwise histograms provide global prefix comparison targets; they do not retain separate source stratum arrays. These checks compute exact central minima, not complete true minimum vectors in the far range. Far-envelope domination, the infinite estimates and the reduction arguments still use the proved mathematics and the source producer. The attributed binary mean theorem remains an import; this reconstruction does not repeat its earlier paper.

The paper proves mean bounds for its full-unit rational-height model. It does not determine the optimal correction, maximizing lengths or a sign classification for all bases with at least three primes. It makes no assertion about the Collatz conjecture. The classical tools and earlier binary theorem are attributed, and no universal priority claim is made.

Data and code availability. All author-generated integer certificates and programs needed for the finite proof dependencies accompany the paper in the repository. The manuscript is licensed under CC BY 4.0; original code, certificates and repository documentation are licensed under MIT. Cited publications retain their own rights and are not bundled.

AI-use disclosure. OpenAI GPT models assisted with mathematical exploration, proof criticism, literature searches, manuscript editing, implementation and reproducibility checks. Several AI review passes shared source exposure. These do not constitute independent human peer review, formal proof verification or an originality certificate. The author retains responsibility for the arguments, source attributions and accompanying artifacts.

A A finite compensation inequality for two-prime moduli

Let $M = p^a r^b$, where $p < r$ are distinct primes and $a, b \geq 1$. Throughout this appendix the average is over all M residues, including zero, and

$$R_M(s) = \min_{\substack{d>0,\ (d,M)=1 \\ A \equiv sd \pmod M}} \max(|A|, d), \qquad \mu(M) = \frac{1}{M} \sum_{s \bmod M} \log R_M(s) - \frac{1}{2} \log M.$$

All logarithms are natural. Write

$$A_0 = \sqrt{M}, \quad D = pr, \quad u = (1 - p^{-1})(1 - r^{-1}),$$
$$\rho = \frac{pr}{(p+1)(r+1)}, \quad \alpha = \zeta(2)^{-1}, \quad \ell = \log 2.$$

$$\kappa = \alpha\rho, \qquad \theta = \alpha(1 - \rho), \qquad \beta = 3 - 2\sqrt{2}.$$

The symbol A_0 denotes a real scale, not a numerator.

A.1 Unit discrepancy and primitive counting

Define

$$U(x) = \lfloor x \rfloor - \lfloor x/p \rfloor - \lfloor x/r \rfloor + \lfloor x/D \rfloor, \qquad e(n) = U(n) - un.$$

Let $h = \max_{0 \leq n < D} e(n)$, $m = \min_{0 \leq n < D} e(n)$, and put

$$K = h - m + u, \qquad W = \max(h, -m + 2u), \qquad H_0 = u + h.$$

Then $h \geq 0$, $m \leq 0$, and $H_0 \geq 1$, the last inequality following from $e(1) = 1 - u$. The real discrepancy lies in $[m - u, h]$. Every closed interval of length v in the positive half-line therefore contains at most $uv + K$ positive D-units: for its integer endpoints $i = \lceil x \rceil$, $j = \lfloor y \rfloor$, use $U(j) - U(i-1)$ and $j - i \leq y - x$. This includes both endpoint atoms and zero-width intervals. Partial summation gives

$$\sum_{\substack{d \leq Q \\ (d,D)=1}} \frac{1}{d} \leq H_0 + u \log Q \quad (Q \geq 1).$$

For $x = n + t$, $0 \leq t < 1$,

$$nU(n) - ux^2 = ne(n) - u(2nt + t^2), \qquad |nU(n) - ux^2| \leq Wx.$$

For the lower bound one uses

$$(-m + 2u)x - \{-nm + u(2nt + t^2)\} = 2un(1 - t) + (-m + 2u)t - ut^2 \geq 0;$$

the upper bound follows from $h \geq 0$.

Let $P(T)$ count pairs $1 \leq A, d \leq T$ with $(A, d) = 1$ and $(d, D) = 1$. For integer $T \geq 1$, cancelling content only at D-units yields

$$P(T) = \sum_{\substack{g \leq T \\ (g,D)=1}} \mu_{\mathrm{Mob}}(g) \lfloor T/g \rfloor U(\lfloor T/g \rfloor).$$

The Euler product

$$u \sum_{(g,D)=1} \frac{\mu_{\mathrm{Mob}}(g)}{g^2} = \frac{u\alpha}{(1 - p^{-2})(1 - r^{-2})} = \kappa$$

and the integer-cutoff bound $\sum_{g>T} g^{-2} \le T^{-1}$ prove

$$|P(T) - \kappa T^2| \le T(b_0 + b_1 \log T), \tag{1}$$

with either of the *coherent* choices

$$(b_0, b_1) = (W + u, W) \quad \text{or} \quad (b_0, b_1) = (WH_0 + u, Wu).$$

Both have $b_0 \ge b_1$. The elementary bounds $h < 2$, $m > -1$ also permit the coarse pair $(b_0, b_1) = (4, 3)$ and $K = 4$: here $|nU(n) - ux^2| \le 3x$, and the unmarked harmonic sum is at most $1 + \log T$.

For later sharpening we need the stronger uniform integer discrepancy

$$-u \le e(n) \le 2u. \tag{2}$$

It suffices to work in one period. At units $e(n) - e(n-1) = 1 - u > 0$, so a minimum occurs at zero or a nonunit. If $n = pt < D$, write $pt = jr + s$, with $1 \le t < r$, $0 \le j \le p-1$, $1 \le s < r$. Then $e(pt) = (s-t)/r$, and

$$t - s = jr - (p-1)t \le \frac{jr - (p-1)}{p} \le \frac{(p-1)(r-1)}{p} = ur.$$

Interchanging the primes treats multiples of r. Reflection gives $e(D-n) = \mathbf{1}_{(n,D)=1} - e(n)$ for $0 < n < D$. At nonunits this implies $e(n) \le u$. At a unit t, the lower bound for $e(t-1)$ implies $e(t) \ge 1 - 2u$; reflection then gives $e(n) \le 2u$. Thus $K \le 4u$, $W \le 3u$. For the second coherent parameter choice,

$$b_0 \le 9u^2 + u, \quad b_1 \le 3u^2, \quad b_0 - b_1 = Wh + u \le 6u^2 + u. \tag{3}$$

A.2 Central collisions with closed endpoints

For integer $T^2 \le M$, let $C(T) = \#\{s : R_M(s) \le T\}$. Every nonzero short witness can be cancelled to a primitive one because its content is a unit modulo M. Two primitive witnesses with the same numerator sign represent different states unless they are identical: a nonzero determinant would be a multiple of M, while its magnitude is $< M$. At $T^2 = M$, equality in this bound would require a denominator T, which is not a unit. Opposite signs can meet only when

$$Ad + Eb = M, \qquad 1 \le A, E, b, d \le T, \quad (bd, D) = 1.$$

The value $2M$ is impossible for the same nonunit-endpoint reason. There is at most one witness of either sign at a given state. Neither a nonzero short witness nor its negative can be the zero state. Consequently

$$C(T) = 1 + 2P(T) - J(T), \tag{4}$$

where $J(T)$ counts the ordered positive/negative collision pairs above; no further factor two is inserted.

A common divisor of b, d divides M and is coprime to M, so $(b, d) = 1$. For fixed b, d, the interval of progression parameters has width at most

$$\frac{T(b+d) - M}{bd} < 1$$

when nonempty. Indeed $T(b+d) - M - bd \le -(T-b)(T-d)$, with equality requiring $M = T^2$ and one nonunit denominator T. Hence at most one collision occurs per ordered denominator

pair. Put $L_T = (2T - M/T)_+$. The deficits $T-b, T-d$ lie in the closed triangle $x+y \le L_T$. If $V(v)$ counts admissible deficits up to v, then $V(v) \le uv + K$. Summing the linear weights as a Stieltjes integral gives

$$J(T) \le \frac{u^2 L_T^2}{2} + 2uKL_T + K^2 \quad (2T^2 > M). \tag{5}$$

There are no collisions for $2T^2 < M$. If $2T^2 = M$, equality forces $b = d = T$; the odd prime divisor of M divides T, so these are not units. Thus $J(T) = 0$ also at this boundary.

A.3 The global far cover

For integer $T^2 \ge 2M$, consider the complete Farey sequence of order T. A unit-denominator center c/d covers the closed interval $[c/d - T/(Md), c/d + T/(Md)]$. Consecutive centers $a/q < c/d$ have $cq - ad = 1$, $(q,d) = 1$, and $q + d > T$. If both denominators are units their intervals overlap. If only one is a unit, an uncovered point forces the other denominator to be $< M/T$.

If both are nonunits, their two prime obstructions are complementary. Hence $q - d$ is a unit modulo D. On the Farey arc the affine residual

$$(q-d)x - (a-c)$$

has values $1/q$ and $1/d$ at its endpoints. If $q, d \ge M/T$, its absolute value multiplied by M is at most T, and the vector difference is a genuine witness of denominator $0 < |q-d| \le T$ on the whole arc. A witness center outside that arc is allowed; no isolation of deleted centers is presumed.

It follows that every uncovered point belongs to an arc incident to a deleted reduced center a/q, $0 < a < q$, with $q < M/T \le T/2$. A complete Farey neighbor c/d has $d > T - q > q$. If d is also a nonunit, replace it by $(c-a)/(d-q)$. Its denominator is a unit, lies in $(T-2q, T]$, and has determinant one with a/q on the same side. If $D \mid q$, the original neighbors are already units and lie in $(T-q, T]$. The uncovered residues are therefore contained in the union of open holes

$$\left(\frac{Ma}{q} - \frac{M - qT}{qb_L}, \frac{Ma}{q} + \frac{M-qT}{qb_R}\right), \qquad b_L, b_R > T - 2q.$$

An open interval of length v contains at most $v+1$ integers. Overlapping holes may be counted repeatedly in an upper bound. Consequently

$$M - C(T) \le \sum_{\substack{q < M/T \\ (q,D)>1}} \varphi(q) \left\{ \frac{2(M-qT)}{q(T-2q)} + 1 \right\}. \tag{6}$$

The $+1$ allowances, positive denominator marks and open endpoints are essential. The displayed cover holds also at $T^2 = 2M$, but a later partial summation must start strictly above that threshold.

A.4 Deleted-center density and its real-cutoff error

Put $A(Q) = \sum_{q \le Q,\,(q,D)>1} \varphi(q)/q$, and define

$$v = 2 - p^{-1} - r^{-1}, \qquad c_0 = \frac{p-1}{p^2} + \frac{r-1}{r^2} - \frac{D-1}{D^2}.$$

Expanding $\varphi(q)/q = \sum_{d \mid q} \mu_{\mathrm{Mob}}(d)/d$ and separating its p, r parts yields the exact identity

$$A(Q) = \sum_{\substack{d \le Q \\ (d,D)=1}} \frac{\mu_{\mathrm{Mob}}(d)}{d} \left\{ (1-p^{-1}) \left\lfloor \frac{Q}{pd} \right\rfloor + (1-r^{-1}) \left\lfloor \frac{Q}{rd} \right\rfloor - (1-D^{-1}) \left\lfloor \frac{Q}{Dd} \right\rfloor \right\}.$$

The linear coefficient in braces is $c_0 Q/d$, and their error is at most v in absolute value. Since $c_0 = (1-p^{-2})(1-r^{-2}) - u$, the leading density is θ. For real $Q \ge 1$, not merely integer Q,

$$Q \sum_{d>Q} d^{-2} \le Q/\lfloor Q \rfloor < 2.$$

Thus

$$|A(Q) - \theta Q| \le E(Q) = E_0 + E_1 \log Q \tag{7}$$

with either $E_0 = v + 2c_0, E_1 = v$, or $E_0 = vH_0 + 2c_0, E_1 = vu$. The coarse admissible choice is $E(Q) = 3 + 2\log Q$, since $c_0 \le 17/36$. For partial summation on $0 \le Q < 1$, take $E(Q) = E_0$: then $A(Q) = 0$, $\theta Q \le \theta < 1/3$, and $E_0 \ge vu \ge 7/18$. A second partial summation gives

$$B(Q) := \sum_{\substack{q \le Q \\ (q,D)>1}} \varphi(q) \le \frac{\theta Q^2}{2} + 2QE(Q). \tag{8}$$

For $T^2 > 2M$, write $Q = M/T$ and $f(x) = (M - Tx)/(T - 2x)$. It decreases from M/T to zero at Q. Equations (6)–(8) imply

$$\frac{M - C(T)}{M} \le V(T/A_0) + \frac{4E(Q)}{T}, \qquad V(t) = \theta F(t/\sqrt{2}) + \frac{\theta}{2t^2},$$

where

$$F(z) = \sum_{n \ge 1} \frac{z^{-2n}}{n(n+1)}, \qquad F(1) = 1.$$

Indeed $2\theta M^{-1} \int_0^Q f(x)\,dx = \theta F(T/(\sqrt{2}A_0))$; the weighted discrepancy costs $2E(Q)/T$, and the integer-hole allowance costs another $2E(Q)/T$. The function V is decreasing, $V(\sqrt{2}) = 5\theta/4$, and termwise integration of its positive series gives

$$\int_{\sqrt{2}}^{\infty} V(t)\frac{dt}{t} = \theta\left\{\frac{\zeta(2) - 1}{2} + \frac{1}{8}\right\} = \frac{1-\rho}{2} - \frac{3\alpha(1-\rho)}{8}. \tag{9}$$

A.5 The complete finite compensation

Denominator one bounds every height by $\max(1, \lfloor M/2 \rfloor)$. The layer-cake identity is

$$\mu(M) = -D_c + P_{\mathrm{mid}} + P_{\mathrm{far}},$$

with

$$D_c = \frac{1}{M}\int_1^{A_0} C(\lfloor x \rfloor)\frac{dx}{x}, \qquad P_{\mathrm{mid}} = \frac{1}{M}\int_{A_0}^{\sqrt{2}A_0} [M - C(\lfloor x \rfloor)]\frac{dx}{x}.$$

The floor lemma

$$\int_1^{A_0} \lfloor x \rfloor^2 \frac{dx}{x} \ge A_0^2/2 - A_0$$

follows by summing $\log(1 + 1/n) \ge 1/n - 1/(2n^2)$ and using $\log((k+t)/k) \ge t/(k+t)$ on the last interval. The residual comparison is at least $1/2 + t - 3t^2/2 \ge 0$ for $0 \le t < 1$. Integration of (1) and (5), retaining the closed collision boundary, gives

$$D_c \ge \kappa - u^2(3/4 - \ell) - \frac{2\kappa + 2(b_0 - b_1) + 2b_1 \log A_0 + 2uK\beta}{A_0} - \frac{K^2\ell}{2A_0^2}.$$

Here the two collision integrals are

$$\frac{1}{M}\int_{A_0/\sqrt{2}}^{A_0} (2x - M/x)^2 \frac{dx}{x} = 3/2 - 2\ell, \qquad \int_{A_0/\sqrt{2}}^{A_0} (2x - M/x)\frac{dx}{x} = A_0\beta.$$

The favorable central zero and favorable primitive-error endpoint $2(b_0-b_1)/A_0^2$ have only been discarded. At $k=\lfloor A_0\rfloor$, $k^2\ge A_0^2-2A_0$. CDF monotonicity throughout the middle interval implies

$$P_{\mathrm{mid}}\le(1-2\kappa+u^2/2)\frac{\ell}{2}+\frac{\ell(4\kappa+2b_0+2b_1\log A_0+2uK)}{2A_0}+\frac{K^2\ell}{2A_0^2}.$$

Start the discrete far sum at

$$L=\lfloor\sqrt{2M}\rfloor+1.$$

The initial interval $[\sqrt{2M},L]$ costs at most $1/(\sqrt{2}A_0)$, including when $\sqrt{2M}$ is an integer. This prevents the false substitution $f(Q)=0$ at $T^2=2M$. For all subsequent terms $T^2>2M$. Use $\log(1+1/T)\le 1/T$, decreasing-sum comparison and $\sum_{T\ge L}T^{-2}\le L^{-1}+L^{-2}$. Equations (9) and (7) then give

$$P_{\mathrm{far}}\le\frac{1-\rho}{2}-\frac{3\alpha(1-\rho)}{8}+\frac{1+5\theta/4+4E(A_0)}{\sqrt{2}A_0}+\frac{2E(A_0)}{A_0^2}.$$

When $Q<1$ the survival is zero; no negative logarithmic error is used. Combining the three bounds proves

$$\mu(M)\le-\Delta(p,r)+\mathcal{E}_{p,r}(A_0),\tag{10}$$

where

$$\Delta=\rho\{\alpha(5/8+\ell)+1/2\}+3\alpha/8-1/2-\ell/2-\tfrac{3}{4}(1-\ell)u^2$$

and

$$\begin{aligned}\mathcal{E}_{p,r}(A_0)=\frac{1}{A_0}\big[&2\kappa+2(b_0-b_1)+2b_1\log A_0+2uK\beta\\&+\tfrac{\ell}{2}(4\kappa+2b_0+2b_1\log A_0+2uK)+(1+5\theta/4+4E(A_0))/\sqrt{2}\big]\\&+\frac{K^2\ell+2E(A_0)}{A_0^2}.\end{aligned}\tag{11}$$

All these estimates hold for arbitrary positive prime valuations. They are inequalities for the original finite average, not limiting-law or monotonicity assertions.

A.6 Uniform leading margins and universal cutoffs

The margin Δ increases with either prime. To check this algebraically put $x=1/p, y=1/r$, $A_*=\alpha(5/8+\ell)+1/2$, $C_*=\frac{3}{4}(1-\ell)$. Multiplication of $\partial_x\Delta$ by $(1+x)/\rho$ gives

$$-A_*+2C_*(1-x)(1+x)^2(1-y)^2(1+y)<0.$$

For $0\le x,y\le 1/2$, its positive term is $<9/8$, whereas $A_*>51/40$, using $\alpha>3/5$ and $\ell>2/3$. Thus both reciprocal-prime derivatives are negative.

The coarse coherent parameters $(b_0,b_1,K)=(4,3,4)$ and $E(A_0)=3+2\log A_0$ give

$$\mathcal{E}_{p,r}(A_0)<\frac{22+15\log A_0}{A_0}+\frac{18+4\log A_0}{A_0^2}.\tag{12}$$

For example, the coefficients of its linear constant and logarithm are bounded by $5167/240<22$ and $141/10<15$; its quadratic constant is <18. At the smallest corner,

$$\Delta(2,3)=\alpha(11/16+\ell/2)-1/3-5\ell/12>2885/521272>1/200.$$

Indeed convexity bounds

$$\zeta(2)<\sum_{n=1}^{5}n^{-2}+\int_{11/2}^{\infty}x^{-2}dx=65159/39600.$$

Insert $\alpha > 39600/65159$ first and $\ell < 7/10$ second. Elementary finite positive exponential or atanh series give

$$3/5 < \alpha < 2/3, \quad 2/3 < \ell < 7/10, \quad \log 3 < 11/10, \quad \log 5 < 17/10.$$

At $A_0 = 2^{16}$, $\log A_0 < 56/5$, and

$$\mathcal{E} < \frac{190}{65536} + \frac{314/5}{65536^2} < 1/300.$$

The positive error terms decrease thereafter. Hence every two-prime modulus $M \geq 2^{32}$ has $\mu(M) < -1/600$.

The sharper three cutoffs follow from the refined parameters, still without finite-height experiments:

$$\begin{aligned} &p = 2,\ r \geq 11,\ M \geq 384^2: && \mu(M) < -1/600, \\ &p = 3,\ r \geq 7,\ M \geq 512^2: && \mu(M) < -1/350, \\ &p \geq 5,\ r \geq 7,\ M \geq 600^2: && \mu(M) < -1/750. \end{aligned} \tag{13}$$

Here are the exact parameter substitutions and endpoint comparisons.

For $p = 2$ the periodic extrema are $h = 2u, m = -u$; they occur at $n = r-2, r+1$. Thus $K = 4u, W = 3u$. For $r \geq 11$, using the second coherent primitive and density choices,

$$u \leq \tfrac{1}{2}, \quad \kappa < \tfrac{4}{9}, \quad \theta < \tfrac{1}{3}, \quad b_0 \leq \tfrac{11}{4}, \quad b_1 \leq \tfrac{3}{4}, \quad b_0 - b_1 \leq 2, \quad K \leq 2, \quad E_0 \leq \tfrac{125}{44}, \quad E_1 \leq \tfrac{3}{4}.$$

Substitution in (11), using $\beta < 1/5$ and $1/\sqrt{2} < 3/4$, gives

$$\mathcal{E} < \frac{363/20 + (171/40)\log A_0}{A_0} + \frac{17/2 + (3/2)\log A_0}{A_0^2}.$$

The unrounded linear constant is $143521/7920 < 363/20$. At $A_0 = 384$, $\log 384 < 6$, so

$$\mathcal{E} < 73/640 + 35/294912 < 23/200.$$

The exact corner after inserting $\alpha > 3/5$, then $\ell > 2/3$, is

$$\Delta(2, 11) > 109/240 - 761/2178 + (157/7260)(2/3) = 3461/29040 > 7/60.$$

This proves the first line of (13).

For $p = 3, r \geq 7$, (2) gives

$$u \leq \tfrac{2}{3}, \quad \kappa < \tfrac{1}{2}, \quad \theta < \tfrac{1}{3}, \quad K \leq \tfrac{8}{3}, \quad b_0 \leq \tfrac{14}{3}, \quad b_1 \leq \tfrac{4}{3}, \quad b_0 - b_1 \leq \tfrac{10}{3}, \quad E_0 \leq \tfrac{250}{63}, \quad E_1 \leq \tfrac{10}{9}.$$

Therefore

$$\mathcal{E} < \frac{27 + 7\log A_0}{A_0} + \frac{13 + (9/4)\log A_0}{A_0^2}.$$

The unrounded linear constant is $133843/5040 < 27$. Since $\log 512 < 63/10$,

$$\mathcal{E}(512) < 711/5120 + 1087/10485760 < 7/50.$$

The corner computation is

$$\Delta(3, 7) > 603/1280 - 1307/3136 + (1087/7840)(2/3) = 27613/188160 > 1/7,$$

which proves the second line.

Finally, at $p = 5, r = 7$, insertion of $\alpha > 3/5$, then $\ell > 2/3$, gives $\Delta(5, 7) > 95869/470400 > 61/300$. The coefficient of ℓ after the first insertion is positive. Since $600 = 2^3 \cdot 3 \cdot 5^2$, $\log 600 < 33/5$, and (12) gives

$$\mathcal{E}(600) < 121/600 + 37/300000 = 60537/300000 < 101/500.$$

The difference $61/300 - 101/500 = 1/750$ proves the third line. All endpoint comparisons are strict rational inequalities, and all error terms decrease above the stated scale.

Coverage boundary. The three cutoffs cover every pair except radicals $6, 10, 14, 15$. They reduce the remaining problem to a bounded domain, but do not themselves certify that domain or those four families. A complete proof of the all-modulus theorem must additionally supply their separate family arguments, the later source-domain reductions and the exact finite-certificate clauses. No sign extrapolation or monotonicity of the true finite means may replace these remaining premises.

B The refined radical-six balance

This argument applies to every M with $\mathrm{rad}(M) = 6$, not just 6^N. Put $a = \sqrt{M}$, $\alpha = \zeta(2)^{-1}$, $\ell = \log 2$, $\theta = \alpha/2$, and $\theta_6 = \alpha/12$. Period six gives

$$-2/3 \le U_6(x) - x/3 \le 2/3, \qquad -1/3 \le U_6(n) - n/3 \le 2/3 \quad (n \in \mathbb{Z}_{\ge 0}).$$

Partial summation yields $\sum_{d\le Q,(d,6)=1} 1/d \le 1 + \log Q/3$. For real $Q \ge 2$ the tail satisfies $Q\sum_{d>Q,(d,6)=1} d^{-2} \le 1/3 + 4/(3Q) \le 1$. For $1 \le Q < 2$, it is $< 2/9$ since $\sum_{d\ge 2,(d,6)=1} d^{-2} = 2\zeta(2)/3 - 1 < 1/9$. The real-cutoff estimate is therefore valid without rounding Q.

For $x = n + t$, $0 \le t < 1$, $|nU_6(n) - x^2/3| \le x$, by the displayed discrepancy. Möbius inversion with only six-unit contents consequently gives

$$|P(T) - \alpha T^2/2| \le (4/3)T + (1/3)T\log T. \tag{B.1}$$

Every closed interval of length h contains at most $h/3 + 4/3$ six-units, including boundary atoms. One proof uses the alternating spacings two and four: $2m + 1$ consecutive units span at least $6m$ and $2m$ units span at least $6m - 4$. For $L_T = (2T - M/T)_+$, the collision triangle in the preceding appendix then gives, when $2T^2 > M$,

$$J(T) \le L_T^2/18 + (8/9)L_T + 16/9. \tag{B.2}$$

At $2T^2 = M$, T is a nonunit, so the putative equality pair $d = f = T$ is forbidden. At $T^2 = M$, T is also a nonunit. Thus the same endpoint conventions apply for unequal valuations.

There is a useful additional gain in the far tail. Define

$$A(Q) = \sum_{\substack{d\le Q\\(d,6)>1}} \varphi(d)/d, \qquad A_6(Q) = \sum_{\substack{d\le Q\\6\mid d}} \varphi(d)/d.$$

Möbius expansion gives the exact identities

$$A(Q) = \sum_{\substack{d\le Q\\(d,6)=1}} \frac{\mu_{\mathrm{Mob}}(d)}{d}\left[\frac{1}{2}\lfloor Q/(2d)\rfloor + \frac{2}{3}\lfloor Q/(3d)\rfloor - \frac{5}{6}\lfloor Q/(6d)\rfloor\right],$$

$$A_6(Q) = \frac{1}{3}\sum_{\substack{d\le Q/6\\(d,6)=1}} \frac{\mu_{\mathrm{Mob}}(d)}{d}\lfloor Q/(6d)\rfloor.$$

The first bracket minus $Q/(3d)$ has period six and lies in $[-2/3, 1/3]$. The harmonic and real tail bounds above then give, for real $Q \ge 1$,

$$|A(Q) - \theta Q| \le E(Q) = 1 + (2/9)\log Q, \quad |A_6(Q) - \theta_6 Q| \le E_6(Q) = 2/3 + (1/9)\log Q. \tag{B.3}$$

For example, decompose a denominator into its powers of two and three and its six-unit part; summing the powers gives geometric series, and the six-unit harmonic and tail estimates above bound the remaining floor errors. Equivalently insert $p = 2, r = 3$ into the exact deleted-density

floor identity in Appendix A, and separately retain denominators divisible by both primes. The resulting coefficients are $\theta = \alpha/2$ and $\theta_6 = \alpha/12$.

For centers divisible by six the neighbor repair costs d, rather than $2d$. With F defined in Section 4, the main survival envelope is therefore

$$V(t) = (\theta - \theta_6)F(t/\sqrt{2}) + 2\theta_6 F(t) + \theta/(2t^2), \quad t \geq \sqrt{2}.$$

In partial summation, the difference between the two rational kernels is unimodal; its maximum is at most $2M^2/T^3$ and its total variation is at most twice this. The extra error is consequently $8E_6(M/T)M/T^3$; the original one is $4E(M/T)/T$. Termwise positive-series integration, or the identity $2\,\mathrm{Li}_2(1/2) = \zeta(2) - \ell^2$, gives

$$\int_{\sqrt{2}}^{\infty} V(t)\frac{dt}{t} = 1/4 - 3\alpha/16 - \alpha(1-\ell)^2/24.$$

The dilogarithm identity follows by differentiating $\mathrm{Li}_2(x) + \mathrm{Li}_2(1-x) = \zeta(2) - \log x \log(1-x)$ and fixing the endpoint.

Repeat the central and middle integrals of the preceding appendix with (B.1)–(B.2), and start the far sum at $\left\lfloor \sqrt{2M} \right\rfloor + 1$. The complete result is

$$\mu(M) \leq -\Delta_6 + \mathcal{E}_6(a),$$

where

$$\Delta_6 = \alpha(11/16 + \ell/2) - 1/3 - 5\ell/12 + \alpha(1-\ell)^2/24 > 1/130$$

and, writing $\beta = 3 - 2\sqrt{2}$,

$$\begin{aligned}\mathcal{E}_6(a) = \frac{1}{a}\big[&\alpha + 2 + (2/3)\log a + (8/9)\beta \\ &+ \ell(2\alpha + 32/9 + (2/3)\log a)/2 \\ &+ (1 + 5\alpha/8 + 4E(a) + (4/3)E_6(a))/\sqrt{2}\big] \\ &+ \{16\ell/9 + 2E(a) + 2E_6(a)\}/a^2.\end{aligned}$$

The margin follows from $\alpha > 39600/65159$, $\ell < 7/10$: the part without the final gain exceeds $2885/521272$, and the gain exceeds $1/440$, whose sum exceeds $1/130$. At $a_0 = 6^{9/2} > 3000$, $\log a_0 < 81/10$; using $\alpha < 2/3, \ell < 7/10, 1/\sqrt{2} < 3/4, \beta < 1/5$ gives

$$\mathcal{E}_6(a_0) < 23/a_0 + 10/a_0^2 < 6901/900000 < 1/130.$$

Every error term decreases for $a \geq a_0$. Hence $\mu(M) < -287/11700000$ whenever $\mathrm{rad}(M) = 6$, $M \geq 6^9$. The eight smaller diagonal moduli $6^1, \ldots, 6^8$ are finite certificates; unequal valuations below the cutoff are handled separately.

C Elementary constant enclosures

All nonrational constants used for cutoff comparisons admit positive series enclosures. In particular

$$\sum_{n=1}^{10} n^{-2} + \frac{1}{11} > \frac{41}{25}, \qquad \zeta(2) < \sum_{n=1}^{5} n^{-2} + \frac{2}{11} = \frac{65159}{39600} < \frac{33}{20}.$$

For the upper bound, convexity of x^{-2} bounds the term at each integer $n \geq 6$ by its integral on $[n - 1/2, n + 1/2]$. The lower bound uses the integral tail after ten terms. Further,

$$e \leq \sum_{n=0}^{6} \frac{1}{n!} + \frac{1}{7!(1 - 1/8)} < \frac{87}{32} < 3.$$

The atanh enclosure in Section 3, applied to rational arguments, proves all logarithmic comparisons with outward rounding. For instance its range reduction gives $2/3 < \log 2 < 7/10,\ \log 3 < 11/10,\ \log 5 < 17/10,\ \log 140 < 5,\ \log 384 < 6,\ \log 512 < 63/10,\ \log 600 < 13/2$. Alternatively, lower partial sums of $\exp x$ prove each upper logarithm comparison. These checks do not rely on floating decimal evaluation.

D The short-length bridges

Here we make explicit the predecessor result at lengths at least two used in Appendix E. This order avoids a circular use of the length-one completion. First the three sixth-power certificates for $B = 10, 12, 14$, the eight base-six certificates, and the coarse cutoffs give every two-prime base at $N \geq 6$, as explained in the family proof. The complete radical families then follow without using any length-two or length-one result.

Outside radicals $6, 10, 14, 15$, the coarse balance admits further corner cutoffs. Monotonicity of its leading margin gives

$$\Delta(2, 11) > 3461/29040 > 1/9,$$
$$\Delta(3, 7) > 27613/188160 > 1/7, \quad \Delta(2, 17) > 1447/10404 > 1/8.$$

At $a = 1296 = 6^4$, using $\log a < 36/5$, the coarse error $(22 + 15 \log a)/a + (18 + 4 \log a)/a^2$ is $< 46813/466560 < 1/9$. At $a = 1024$, using $\log a < 7$, it is $< 65047/524288 < 1/7$ and $< 1/8$. These comparisons and decreasing errors give:

remaining prime factors	sufficient M
$p = 2,\ r = 11$ or 13	6^8
$p = 2,\ r \geq 17$	2^{20}
$p \geq 3,\ r \geq 7$	2^{20}.

(D.1)

In particular all lengths $N \geq 5$ are covered. For length four the complete remaining bases are $21, 22, 26$; their three finite certificates prove all lengths $N \geq 4$.

For length three define the conservative domain of bases $B = p^a r^b$ outside the four complete radicals by $B^3 < 6^8$ if $p = 2, r = 11, 13$, and $B^3 < 2^{20}$ otherwise. Remove bases already covered by reindexing at length at least four. This gives exactly the following 35 bases:

$$21, 22, 26, 33, 34, 35, 38, 39, 44, 46, 51, 52, 55, 57, 58, 62, 63, 65, 68, 69,$$
$$74, 76, 77, 82, 85, 86, 87, 88, 91, 92, 93, 94, 95, 99, 104.$$

The retained domain generator also gives their unique prime valuations. Every one has a negative certificate, proving all lengths $N \geq 3$. The numerical limits behind the enumeration are $B \leq 118$ in the first category and $B \leq 101$ in the others.

For length two enumerate every integer base $B \leq 1295$ with exactly two prime factors outside the four radicals. Keep

$$B < 1296 \quad (p = 2, r = 11, 13), \qquad B < 1024 \quad \text{otherwise},$$

and remove $B = 441, 484, 676$, which are respectively the squares of the already certified bases $21, 22, 26$. Any other higher-power reindexing in this range is either in a complete radical family or above its analytic cutoff. There are exactly 454 retained bases, split into 256 odd bases, 183 bases with $p = 2, r \geq 17$, and 15 with $p = 2, r = 11, 13$. Their physical modulus is $M = B^2$; they are not 454 arbitrary two-prime moduli. The parameterized balance with the actual periodic discrepancy and outward constant enclosures pays 355 of them. The other 99 have negative histogram/log certificates constructed by Section 3. This proves every two-prime

base at length two. All lengths $N \geq 2$ are now proved before the length-one residual inventory is formed.

The arithmetic in the analytic filters is also fully finite. For $\zeta(2)$ use

$$\sum_{j=1}^{K} j^{-2} + 1/(K+1) < \zeta(2) < \sum_{j=1}^{K} j^{-2} + 1/K,$$

invert outward to enclose α, and enclose logarithms by the atanh series. Square roots have rational enclosures obtained by integer square roots. The length-two filter uses $K = 128$ for the zeta sum; the companion records the root scale 10^{15}, logarithmic precision and any outward dyadic grid separately. A row is paid only if its entire upper enclosure is strictly negative; an inconclusive enclosure remains in the finite certificate set. The later length-one five-bound filter is evaluated in the same outward manner. A rounded decimal sign is never a filter.

Bridge	Rows	Kind of retained evidence
Odd prime powers	21	exact heights and integer Jensen sums
Base six, $N = 1, \ldots, 8$	8	upper histograms and integer log sums
Length six, bases $10, 12, 14$	3	upper histograms and integer log sums
Squarefree semiprime prefix	4596	digests/signs; reconstruction code
Offdiagonal radical six	143	upper histograms and log ledger
Radical ten	97	upper histograms and log ledger
Radicals fourteen/fifteen	132	upper histograms and log ledger
Length four	3	upper histograms and log ledger
Length three	35	upper histograms and log ledger
Length two residual	99	upper histograms and log ledger
Final length one	4410	upper histograms and log ledger

Table 2: Finite proof dependencies. Domains are specified above and in the companion. Rows with different proof roles may overlap and are not to be added as a count of distinct moduli.

E Effective reduction and finite completion for two prime factors

Write

$$M = p^a q^b, \qquad p < q \text{ distinct primes}, \qquad a, b \geq 1, \qquad \mu(M) = \frac{1}{M} \sum_{s=0}^{M-1} \log R_M(s) - \frac{1}{2} \log M.$$

The minimum defining R_M uses positive denominators coprime to the whole M, and zero is one of the M equally weighted states. All logarithms in this appendix are natural. The reductions below apply to the actual valuations a, b; they do not compare means on different moduli. We write μ_{Mob} for the Möbius function, to distinguish it from the centered mean $\mu(M)$.

E.1 Exact discrepancy parameters and admissible errors

Put

$$D = pq, \quad u = \frac{(p-1)(q-1)}{pq}, \quad \rho = \frac{pq}{(p+1)(q+1)},$$

$$\alpha = \frac{1}{\zeta(2)}, \quad \ell = \log 2, \quad \kappa = \alpha\rho, \quad \theta = \alpha(1-\rho), \quad \beta = 3 - 2\sqrt{2}.$$

For integer $n \geq 0$, let

$$U_D(n) = n - \lfloor n/p \rfloor - \lfloor n/q \rfloor + \lfloor n/D \rfloor, \qquad e(n) = U_D(n) - un.$$

The function e is D-periodic and

$$e(n) - e(n-1) = \mathbf{1}_{(n,D)=1} - u, \qquad e(D-n) = \mathbf{1}_{(n,D)=1} - e(n) \quad (0 < n < D).$$

In particular,

$$-u \leq e(n) \leq 2u. \tag{14}$$

Indeed a minimum occurs at zero or a nonunit. For $n = pi < D$, write $pi = tq + r$, $1 \leq r < q$, $0 \leq t < p$. Then

$$e(pi) = \frac{r-i}{q}, \qquad i - r = tq - (p-1)i \leq \frac{tq - (p-1)}{p} \leq \frac{(p-1)(q-1)}{p} = uq.$$

The other nonunit family is treated with the primes interchanged. Reflection bounds a nonunit above by u. For a unit, its reflected unit follows a step of size $1-u$ from a value at least $-u$; reflection then bounds the original value above by $2u$.

Let σ be the maximum of $e(n)$ over nonunits in $0 \leq n < D$, including $n = 0$. The minimum over the whole period is $-\sigma$, and its maximum is $u + \sigma$: a maximum is immediately before a nonunit step, including the wraparound predecessor $D-1$. An exact short formula, not a scan of all D residues, is

$$\begin{aligned} i_t &= \min\left(q-1, \left\lfloor \frac{q(t+1)-1}{p} \right\rfloor\right), && 0 \leq t < p, \\ \sigma &= \max\left(0, \max_{0 \leq t < p} \frac{(p-1)i_t - qt}{q}, \max_{1 \leq j < p} \frac{(qj \bmod p) - j}{p}\right). \end{aligned} \tag{15}$$

To prove this, the nonunits are pi, $1 \leq i < q$, and qj, $1 \leq j < p$. On the block $t = \lfloor pi/q \rfloor$, the first numerator is increasing in i, so its maximum is at i_t. Every such block is nonempty since $q > p$; the minus one in (15) preserves its strict upper endpoint. There are at most $2p-1$ nonzero candidates.

Define

$$K = 2u + 2\sigma, \qquad W = 2u + \sigma.$$

The real discrepancy lies between $-\sigma - u$ and $u + \sigma$. Consequently a closed interval of length h contains at most $uh + K$ positive D-units, with both boundary atoms included. Also

$$\sum_{\substack{d \leq Q \\ (d,D)=1}} \frac{1}{d} \leq u + (u + \sigma) + u \log Q \qquad (Q \geq 1).$$

For $x = n + t$, $n = \lfloor x \rfloor$, $0 \leq t < 1$, the same discrepancy bounds give $\left|nU_D(n) - ux^2\right| \leq Wx$. Möbius inversion and its integer-cutoff tail therefore yield

$$\left|P(T) - \kappa T^2\right| \leq T(b_0 + b_1 \log T), \qquad P(T) = \sum_{\substack{d \leq T \\ (d,D)=1}} \mu_{\mathrm{Mob}}(d) \lfloor T/d \rfloor U_D(\lfloor T/d \rfloor), \tag{16}$$

with either admissible pair

$$(b_0, b_1) = (3u + \sigma,\ 2u + \sigma), \tag{17}$$

$$(b_0, b_1) = ((2u + \sigma)^2 + u,\ u(2u + \sigma)). \tag{18}$$

The coefficient follows from

$$u \sum_{(d,D)=1} \frac{\mu_{\mathrm{Mob}}(d)}{d^2} = \frac{u\alpha}{(1-p^{-2})(1-q^{-2})} = \kappa.$$

For the first pair use $\sum_{d\le T} d^{-1} \le 1 + \log T$; for the second use the displayed unit harmonic bound. The omitted series costs at most uT. Thus $b_0 - b_1$ is respectively u and $(2u+\sigma)(u+\sigma)+u$, both nonnegative. The independent coarse choice $(b_0, b_1, K) = (4, 3, 4)$ is also valid.

For deleted denominators, set

$$A(Q) = \sum_{\substack{h\le Q \\ (h,D)>1}} \frac{\varphi(h)}{h}, \quad v = 2 - \frac{1}{p} - \frac{1}{q}, \quad c_0 = \frac{p-1}{p^2} + \frac{q-1}{q^2} - \frac{D-1}{D^2}.$$

Decomposing squarefree totient divisors gives the exact identity

$$A(Q) = \sum_{\substack{d\le Q \\ (d,D)=1}} \frac{\mu_{\mathrm{Mob}}(d)}{d} \left[(1-1/p) \left\lfloor \frac{Q}{pd} \right\rfloor + (1-1/q) \left\lfloor \frac{Q}{qd} \right\rfloor - (1-1/D) \left\lfloor \frac{Q}{Dd} \right\rfloor \right].$$

The full sum's leading coefficient is θ. The bracket's linear coefficient is $c_0 Q/d$, and its error is at most v; for a real cutoff, $Q \sum_{d>Q} d^{-2} < 2$, not merely one. Hence either of the complete errors

$$E_{\mathrm{r}}(x) = v(2u+\sigma) + 2c_0 + vu \log x, \qquad E_{\mathrm{u}}(x) = 3 + 2\log x \tag{19}$$

pays $|A(Q) - \theta Q|$ for real $Q \ge 1$. The refined constant also bounds the below-one partial-summation interval: $\theta < 1/3$, whereas $vu \ge 7/18$. In particular,

$$\sum_{\substack{h\le Q \\ (h,D)>1}} \varphi(h) \le \frac{\theta Q^2}{2} + 2QE(Q).$$

E.2 Finite balance, closed endpoints and the four simultaneous gains

For every integer $T^2 \le M$, the original central count satisfies

$$C_M(T) = 1 + 2P(T) - J_M(T).$$

Reduction of common content is legitimate because it divides a full-unit denominator. Same-sign primitive images are injective. Opposite-sign images meet precisely at $Ad + Eb = M$, with positive $A, E, b, d \le T$. Thus $b + d \ge M/T$, with equality retained, and $(b, d) = 1$. For a fixed ordered denominator pair the allowed phase interval has width at most $[T(b+d) - M]/(bd) \le 1$; equality would require $M = T^2$ and a denominator T, which is nonunit at a square modulus. Putting $L_T = (2T - M/T)_+$, the closed interval unit bound gives

$$J_M(T) \le \frac{u^2 L_T^2}{2} + 2uKL_T + K^2 \quad (2T^2 > M).$$

If $2T^2 = M$, equality would force $b = d = T$; the odd prime divisor of M divides T, so the collision is absent. Thus $J_M(T) = 0$ also at that boundary.

Here is the far-cover argument used in the finite balance. For $T^2 > 2M$, full Farey neighbors of order T have coprime denominators and sum greater than T. Two retained neighbors cover their intervening arc. If both are deleted, their prime obstructions are complementary, so their denominator difference is a full unit; when both denominators are at least M/T, that difference witness covers the arc. An uncovered point must therefore touch a deleted center

c/h with $h < M/T < T/2$. Its complete neighbors have denominators greater than $T - h$; subtracting c/h once repairs a bad neighbor and gives a determinant-one unit neighbor of denominator greater than $T - 2h$. When $D \mid h$, the complete neighbors were already units. The corresponding open hole has length at most

$$\frac{2(M - hT)}{h(T - j_h h)}, \qquad j_h = \begin{cases} 1, & D \mid h, \\ 2, & D \nmid h. \end{cases}$$

One additional integer per reduced center pays rounding; overlaps only enlarge the upper bound. Therefore

$$M - C_M(T) \le \sum_{\substack{2 \le h < M/T \\ (h,D)>1}} \varphi(h) + \sum_{\substack{2 \le h < M/T \\ (h,D)>1}} \frac{2\varphi(h)(M - hT)}{h(T - j_h h)}. \tag{20}$$

The denominators are strictly positive. No center phase or exact tail distribution is needed.

Write $x = \sqrt{M}$, $k = \lfloor x \rfloor$, $t = x - k$. For any compatible primitive pair and deleted error E, define

$$\Delta = \rho\{\alpha(5/8 + \ell) + 1/2\} + 3\alpha/8 - 1/2 - \ell/2 - \tfrac{3}{4}(1 - \ell)u^2, \tag{21}$$

$$\begin{aligned} \mathcal{E}_{b,E}(x) = {} & \frac{2\kappa + 2(b_0 - b_1) + 2b_1 \log x + 2uK\beta}{x} \\ & + \frac{\ell}{2x}(4\kappa + 2b_0 + 2b_1 \log x + 2uK) \\ & + \frac{1 + 5\theta/4 + 4E(x)}{\sqrt{2}x} + \frac{K^2\ell + 2E(x)}{M}. \end{aligned} \tag{22}$$

$$\delta_x = \frac{1 + 2t - 3t^2}{2} + \frac{t^3}{x}, \qquad G_x = 2x(1 - t) + t^2 = 2x - M + k^2,$$

$$\mathcal{G}_b(x) = \frac{\ell\kappa G_x + 2\kappa\delta_x + 2(b_0 - b_1) + \log x}{M}. \tag{23}$$

Then the strengthened finite balance is

$$\mu(M) \le -\Delta + \mathcal{E}_{b,E}(x) - \mathcal{G}_b(x). \tag{24}$$

For completeness, its integration and rounding costs can be checked directly. The centered mean is minus the central CDF integral up to x, plus the survival integrals over $[x, \sqrt{2}x]$ and $[\sqrt{2}x, \infty)$. The collision integrals are

$$\frac{1}{M}\int_{x/\sqrt{2}}^{x} (2y - M/y)^2 \frac{dy}{y} = \frac{3}{2} - 2\ell, \qquad \int_{x/\sqrt{2}}^{x} (2y - M/y)\frac{dy}{y} = x\beta.$$

The central credit and monotone middle plateau, before retaining gains, are respectively at least

$$\kappa - u^2(3/4 - \ell) - \frac{2\kappa + 2(b_0 - b_1) + 2b_1 \log x + 2uK\beta}{x} - \frac{K^2\ell}{2M}$$

and at most

$$(1 - 2\kappa + u^2/2)\frac{\ell}{2} + \frac{\ell(4\kappa + 2b_0 + 2b_1 \log x + 2uK)}{2x} + \frac{K^2\ell}{2M}.$$

For the far sum use $h < M/T$, partial summation of A, and

$$F(z) = \sum_{n \ge 1} \frac{z^{-2n}}{n(n+1)}, \qquad F(1) = 1.$$

The leading normalized survival is bounded by $\theta F(T/(\sqrt{2}x)) + \theta x^2/(2T^2)$. Its logarithmic integral is $(1-\rho)/2 - 3\theta/8$. The first integer far threshold is $\lfloor\sqrt{2M}\rfloor + 1$; the initial interval costs at most $1/(\sqrt{2}x)$, including equality at an integer $\sqrt{2M}$. The first summand and discrete error add $(5\theta/4 + 4E(x))/(\sqrt{2}x) + 2E(x)/M$. These terms give (21)–(22).

The four gains in (23) affect distinct terms. Indeed

$$\int_1^x \lfloor y\rfloor^2 \frac{dy}{y} \geq \frac{(k-1)^2}{2} + \frac{k^2 t}{x} = \frac{x^2}{2} - x + \delta_x.$$

Here $\log(1+1/j) \geq 1/j - 1/(2j^2)$ pays the complete intervals and $\log(x/k) \geq t/x$ pays the last interval. Exact primitive-error integration restores $2(b_0 - b_1)/M$; the isolated zero restores $(\log x)/M$. Retaining the exact endpoint k^2, instead of $M - 2x$, restores $\ell\kappa G_x/M$ on the middle plateau. Collision and far penalties are unchanged, so all four improvements coexist. At a square, $\delta_x = 1/2$, $G_x = 2x$. They must remain paired with their own b_0, b_1.

For the coarse pair $(4, 3, 4)$, the error in (22) is strictly below

$$\mathcal{E}_{\mathrm{c}}(x) = \frac{22 + 15\log x}{x} + \frac{18 + 4\log x}{M}.$$

This follows from $2\kappa \leq 4/3$, $\theta \leq 1/3$, $u \leq 1$, $\ell < 7/10$, $\beta < 1/5$, $1/\sqrt{2} < 3/4$: the linear constant is at most $5167/240 < 22$, its logarithmic coefficient is at most $141/10 < 15$, and the quadratic constant is less than 18. Thus the following are five separately valid, complete upper bounds:

$$-\Delta + \mathcal{E}_{\mathrm{c}} - \mathcal{G}_{(4,3)}, \qquad -\Delta + \mathcal{E}_{b,E} - \mathcal{G}_b, \quad b \in \{(17), (18)\},\ E \in \{E_{\mathrm{r}}, E_{\mathrm{u}}\}.$$

Taking their minimum is legitimate; moving a gain from one bound to another is not.

E.3 Uniform modulus cutoffs

The leading credit increases with either prime parameter. Put $z = 1/p, w = 1/q$, $A_0 = \alpha(5/8 + \ell) + 1/2$, $C_0 = 3(1-\ell)/4$. Multiplying its z-derivative by $(1+z)/\rho > 0$ gives

$$-A_0 + 2C_0(1-z)(1+z)^2(1-w)^2(1+w) < 0.$$

Indeed the two factors are at most $9/4$ and 1, respectively, $2C_0 < 1/2$, and $A_0 > 51/40 > 9/8$, using $\alpha > 3/5$ and $2/3 < \ell < 7/10$. This is monotonicity of the bound, not of the true mean.

Since $\sigma \leq u$, the second primitive pair has

$$b_0 \leq 9u^2 + u, \quad b_1 \leq 3u^2, \quad b_0 - b_1 \leq 6u^2 + u, \quad K \leq 4u.$$

For $p = 2$, reflection gives the exact values $\sigma = u$, $K = 4u$, $W = 3u$. Using the refined deleted error in (22), elementary substitution gives the following bounds and rational endpoint witnesses:

prime range	error upper	x_0	leading credit lower
$p = 2,\ q \geq 11$	$\dfrac{363/20 + (171/40)\log x}{x} + \dfrac{17/2 + (3/2)\log x}{x^2}$	384	$\dfrac{3461}{29040} > \dfrac{7}{60}$
$p = 3,\ q \geq 7$	$\dfrac{27 + 7\log x}{x} + \dfrac{13 + (9/4)\log x}{x^2}$	512	$\dfrac{27613}{188160} > \dfrac{1}{7}$
$p \geq 5,\ q \geq 7$	$\mathcal{E}_{\mathrm{c}}(x)$	600	$\dfrac{95869}{470400} > \dfrac{61}{300}$

For the first two rows use respectively

$$c_0 \leq 13/44,\ E_{\mathrm{r},0} \leq 125/44,\ E_{\mathrm{r},1} \leq 3/4, \quad c_0 \leq 20/63,\ E_{\mathrm{r},0} \leq 250/63,\ E_{\mathrm{r},1} \leq 10/9.$$

For the second row $u \le 2/3$, $\kappa < 1/2$, $K \le 8/3$, $b_0 \le 14/3$, $b_1 \le 4/3$, $b_0 - b_1 \le 10/3$. The first row uses $u \le 1/2$, $\kappa < 4/9$, $K \le 2$, $b_0 \le 11/4$, $b_1 \le 3/4$, $b_0 - b_1 \le 2$. The resulting unrounded linear constants are $143521/7920 < 363/20$ and $133843/5040 < 27$. At the three endpoints,

$$\begin{aligned}\mathcal{E}(384) &< 73/640 + 35/294912 < 23/200,\\ \mathcal{E}(512) &< 711/5120 + 1087/10485760 < 7/50,\\ \mathcal{E}_{\mathrm{c}}(600) &< 60537/300000 < 101/500.\end{aligned}$$

Use $\log 384 < 6$, $\log 512 < 63/10$, $\log 600 < 33/5$; these follow from the positive exponential sums giving $\log 2 < 7/10, \log 3 < 11/10, \log 5 < 17/10$. The constant and logarithmic error terms decrease thereafter. It follows, for arbitrary positive valuations, that

$$\begin{aligned}p = 2,\ q \ge 11,\ M \ge 147456: &\quad \mu(M) < -1/600,\\ p = 3,\ q \ge 7,\ M \ge 262144: &\quad \mu(M) < -1/350,\\ p \ge 5,\ q \ge 7,\ M \ge 360000: &\quad \mu(M) < -1/750.\end{aligned} \tag{25}$$

Equality at these sufficient cutoffs is already covered.

E.4 The larger prime to the first power

Suppose $M = Kq$, $K = p^a$, $q > K$, $M \ge 19600$. In the single-p-mark model, all central denominators are smaller than q and are therefore also full units. Primitive cancellation and the collision-free range through $\lfloor\sqrt{M/2}\rfloor$ transfer unchanged from the single-mark proof. In its Farey tail, adjacent denominators cannot both be multiples of p; its deleted center bracket has denominators greater than $T - h$. Thus the odd-prime estimates of Section 4, and the binary calculation given in the following squarefree subsection, transfer to this modulus without a prime-power assumption:

p	leading credit	error at $x = \sqrt{M}$
2	1/12	$(33/8 + (12/11)\log x)/x + 1/(4x^2)$
3	41/486	$(107/27 + (16/11)\log x)/x + 1/(3x^2)$
≥ 5	91/800	$(151/40 + (24/11)\log x)/x + 1/(2x^2)$

At $x = 140$, $\log 140 < 5$ bounds the errors by quantities strictly below $7/100, 82/1000, 106/1000$. Explicit upper fractions are

$$59021/862400, \quad 467279/5821200, \quad 22619/215600.$$

All three errors decrease for $x \ge 140$. Their strict single-mark margins are respectively greater than $1/75, 287/121500, 31/4000$.

Below real height q, single-mark and full-unit events coincide. For a complete reduced lattice basis, the two-prime short-basis argument gives $R_M \le \sqrt{2}M/\lambda_1$. The all-state incidence bound $\Pr(\lambda_1 \le Y) \le 3Y^2/M$ therefore implies $\Pr(R_M > X) \le \min(1, 6M/X^2)$. If $q \ge 6K$, the extra full-unit logarithmic cost is at most

$$\int_q^\infty \frac{6M}{X^2}\frac{dX}{X} = \frac{3K}{q}.$$

Consequently

$$b = 1, \quad M \ge 19600, \quad q \ge c(p)p^a \quad\Longrightarrow\quad \mu(M) < 0, \qquad c(p) = \begin{cases}225, & p = 2,\\ 1300, & p = 3,\\ 400, & p \ge 5.\end{cases} \tag{26}$$

These ratios imply $q > K$ and $q \ge 6K$. The comparisons are $3/225 = 1/75$, $287/121500 - 3/1300 = 43/789750 > 0$, and $31/4000 - 3/400 = 1/4000 > 0$. The first equality still pays a strict result because its single-mark margin is strict. The proof has $b = 1$; it is not a radical-only rule and cannot be applied with the prime roles reversed.

E.5 Squarefree semiprimes: zero mark loss and the complete prefix

Let $R_{p;M}$ denote the minimum with only $p \nmid d$. Separating witnesses according to whether $q \mid d$ gives exactly

$$R_{p;M}(s) = \min\{R_M(s),\ qR_{p;M/q}(s \bmod M/q)\}.$$

If $q \mid d$, the congruence forces $q \mid A$, and division by q gives the displayed child congruence; conversely multiplying a child witness by q gives a sole-p witness. No arbitrary nonunit content is cancelled. At $M = pq$, the circle-gap cap proved below is $R_{pq} \le q$, while the second term is at least q. Consequently $R_{pq} = R_{p;pq}$ pointwise. Since $\sqrt{pq} < q$, the single-mark transfer of the preceding subsection applies, with NO mark-loss penalty. Its odd-prime estimates give $\mu(pq) < -1/500$ for $p \ge 3$, $pq \ge 19600$.

For the even exterior we spell out the smaller sufficient threshold. The binary single-mark calculation below is valid more generally for $M = 2^a q$, $q > 2^a$, with $x = \sqrt{M} < q$: all its central denominators are full units, and the sole-two-mark Farey tail depends only on the excluded even denominators. In particular it applies at $M = 2q$. The divisor union bound on pairs with odd denominator gives

$$P_2(T) \ge [1 - 3\zeta(2)/8]T^2 - \tfrac{1}{2}T\log T \ge \tfrac{3}{8}T^2 - \tfrac{1}{2}T\log T.$$

Indeed there are at least $T^2/2$ raw pairs; remove every odd content $h \ge 3$, bounding its count by $T^2/(2h^2) + T/(2h)$. The sums are bounded by $\sum_{h\ge3,\ h\ \mathrm{odd}} h^{-2} = 3\zeta(2)/4 - 1$ and $\sum_{3\le h\le T,\ h\ \mathrm{odd}} h^{-1} \le \log T$. Here $\zeta(2) < 5/3$. For $T \le K = \lfloor x/\sqrt{2}\rfloor$, $x = \sqrt{M}$, there are no collisions, and hence $C_M(T) \ge cT^2 - dT\log T$ with $c = 3/4, d = 1$. The quadratic floor integral and monotone plateau give

$$\frac{1}{M}\int_1^x C_M(\lfloor y\rfloor)\frac{dy}{y} \ge c\frac{K^2}{x^2}\left(\frac{1}{2} + \log\frac{x}{K}\right) - \frac{2cK}{x^2} - d\frac{K\log K}{x^2}\left(1 + \log\frac{x}{K}\right).$$

This follows by integrating to K, using $\int_1^K \lfloor y\rfloor^2 dy/y \ge K^2/2 - K$, bounding the primitive-error integral by $K\log K$, and retaining the plateau $C_M(K)\log(x/K)$. The displayed $2cK$ is a safe enlargement of the floor loss. For $x \ge 115$,

$$K/x > 7/10 - 1/115 > 11/16, \quad K/x < 3/4,$$
$$K^2/x^2 \ge 1/2 - \sqrt{2}/x, \quad \ell/2 \le \log(x/K) \le 5/11.$$

With $2/3 < \ell < 1$ and $\sqrt{2} < 3/2$, this yields central credit $5/16 - [9/4 + (12/11)\log x]/x$. The single-mark Farey tail has majorant

$$V(z) = \tfrac{1}{2}F(z) + \frac{1}{8z^2}, \quad z \ge 1, \qquad V(1) = 5/8, \qquad \int_1^\infty V(z)\frac{dz}{z} = \frac{\zeta(2)-1}{4} + \frac{1}{16} < \frac{11}{48}.$$

The initial interval costs $1/x$, the first summand $5/(8x)$, and the integer-hole rounding costs $1/(4x) + 1/(4x^2)$. Subtracting the central credit gives, for every $x \ge 115$,

$$\mu(2q) \le -\frac{1}{12} + \frac{33/8 + (12/11)\log x}{x} + \frac{1}{4x^2}.$$

The error decreases there. Since $\log 115 < 5$, its remaining strict margin at $x = 115$ is

$$\frac{1}{12} - \frac{843}{10120} - \frac{1}{52900} = \frac{1127}{80302200} > 0.$$

Thus every odd prime $q \ge 6613$ lies in this analytic exterior: $2q \ge 13226 > 115^2$.

The complete prefix is exactly

$$\{2q : 3 \le q < 6613,\ q \text{ prime}\} \ \cup\ \{pq : 3 \le p < q,\ p, q \text{ prime},\ pq < 19600\}.$$

It has 853 even and 3743 odd moduli, 4596 in total. Every row has a negative outward-log certificate obtained from a legal full-unit upper histogram, capped at the proved q; the pointwise construction is proved in Section 3.2, and the logarithmic certificate inequality is given below. This finite prefix plus the analytic exterior proves every squarefree-semiprime mean strictly negative, without any transfer to higher valuations or any sampling argument.

E.6 Complete small-radical families and their finite witnesses

For clarity, the earlier complete families have an explicit infinite part and a separately retained finite part. The uniform coarse balance gives the cutoffs 2^{22} for odd prime pairs and $9\,000\,000$ for $p = 2, q \ge 5$: at $x = 2048$ its error is less than $7/100$ and $\Delta(3,5) > 1691/14400 > 1/10$; at $x = 3000$ its error is less than $1/20 - 2026/937500$ and $\Delta(2,5) > 263/4500 > 1/20$.

The separately proved refined radical-six finite balance gives $\mu(M) < -287/11700000$ for every modulus of radical six with $M \ge 6^9$. It retains the shorter repaired bracket at centers divisible by both primes, and applies to unequal valuations, with nonunit central equality endpoints and a strictly started far sum. This preceding radical-six lemma, rather than an asymptotic sign or the coarse error above, is used for that one infinite family.

The following finite witness table completes the families. Each entry means that every original modulus in the specified set has a retained negative outward-log upper certificate, not that its true minimum distribution was fully enumerated.

family	finite set below its analytic cutoff	rows
6^n	$1 \le n \le 8$	8
2^a3^b	$M < 6^9,\ a \ne b,\ \gcd(a,b) < 6$	143
2^a5^b	$M < 9\,000\,000,\ (a,b) \ne (1,1),\ \gcd(a,b) < 6$	97
2^a7^b	$M < 9\,000\,000,\ (a,b) \ne (1,1),\ \gcd(a,b) < 6$	78
3^a5^b	$M < 2^{22},\ (a,b) \ne (1,1),\ \gcd(a,b) < 6$	54
B^6	$B \in \{10, 12, 14\}$	3

These rows are not summed into a single disjoint inventory: their roles in the dependency proof differ. The first row plus the six-radical cutoff proves the base-six all-length statement. Every two-prime base at length at least six is then covered: at length six, bases at least fifteen have modulus greater than 6^9, base six is already covered, and the other bases below fifteen are exactly $10, 12, 14$, the last row. For lengths at least seven, base ten reaches its even cutoff, and bases at least twelve reach 6^9. Thus a modulus with $\gcd(a,b) \ge 6$ is covered by reindexing at that same modulus. The middle four rows, the squarefree-semiprime theorem, and their respective large-modulus cutoffs now exhaust every valuation in radicals $6, 10, 14, 15$.

E.7 Certificate arithmetic and exhaustive final partition

The finite certificates use disjoint original gcd strata. For $g \mid M$, $g < M$, put $m = M/g$ and

$$P_g(T) = \sum_{\substack{e \le T/g \\ (e,D)=1}} \mu_{\mathrm{Mob}}(e) U_m(\lfloor T/(ge) \rfloor) U_D(\lfloor T/e \rfloor).$$

Through $\lfloor \sqrt{M} \rfloor$, the exact stratum CDF is $2P_g(T) - J_g(T)$, where ordered closed collision phases $Af + Ed = M$ are assigned to $g = \gcd(A, M)$. A centered inverse modulo m, of magnitude $h \le H = \lfloor (m-1)/2 \rfloor$, supplies numerator $\pm g$. If one prime v is missing from m,

repair a denominator divisible by v to $m-h$, reversing the numerator sign. The repair remains a unit of the original M. For $T \geq g$ its CDF is $2U_D(\min(T,H))$, with the extra pure-child term

$$2\max\{0, V_v(H) - V_v(\max(0, m-T-1))\}, \qquad V_v(z) = \lfloor z/v \rfloor - \lfloor z/D \rfloor.$$

Denominator one independently contributes $2U_m(\min(\lfloor T/g \rfloor, H))$. At $m=2$ each distribution is one atom at height g, not a doubled pair; both are zero for $T < g$. The circle-gap argument with p multiples of a state gives the global cap M/p: the index difference is less than p, hence full-unit, and its centered residue is at most M/p. The state M/p attains this cap. Thus each stratum is completely covered at

$$K_g = \min\{M/p,\ g\lfloor m/2 \rfloor,\ \max(g, D_m)\}.$$

$$D_m = \begin{cases} 1, & m = 2, \\ H, & \mathrm{rad}(m) = D, \\ \max(H, m-v), & m \text{ is a pure-prime power missing } v. \end{cases}$$

The earliest radical-six stratum batch used the wider cap $\lfloor m r_m/2 \rfloor$, with $r_m = 1$ for a mixed child and $r_m = v$ for a pure child missing prime v. This is also safe: in the mixed case it is at least H; in the pure case, $v \geq 2$ makes it at least m, hence at least every repaired denominator $m-h$. The singleton $m=2$ remains one explicit atom at height g. Later batches use the refined D_m above. These safe recipes are distinguished in the provenance map.

Take the maximum of these credits inside a stratum, sum only over disjoint strata, add zero once, and combine by maximum with the global far credit from (20). On an ordered calendar containing all integers through $\lfloor \sqrt{M} \rfloor$, successive $\lceil 257T/256 \rceil$ thereafter, all divisors/caps, the first strict far threshold, and M/p, cumulative maxima give integer lower CDF values L_i ending at M. The histogram $h_i = L_i - L_{i-1}$ dominates the true height even between knots, by CDF monotonicity. Therefore

$$\mu(M) \leq \frac{2\sum_i h_i u(t_i) - M l(M)}{2M\,2^{64}} = \frac{Q_M}{2M\,2^{64}}, \quad l(n) \leq 2^{64}\log n \leq u(n).$$

This doubled denominator is unchanged for odd M. The enclosures follow from binary range reduction and

$$0 \leq \log y - 2\sum_{j=0}^{K-1} \frac{z^{2j+1}}{2j+1} \leq \frac{2z^{2K+1}}{(2K+1)(1-z^2)}, \qquad z = \frac{y-1}{y+1}, \quad 1 \leq y \leq 2.$$

The finite assertion is $Q_M < 0$ on every row of each declared set. Large intermediate central counts rely on the exact formula and its generator; a retained checker is not described as a complete second implementation of every central count or of the true far survival.

We now use the previously proved squarefree-semiprime theorem and the two-prime all-length theorem for lengths at least two. If $e = \gcd(a,b) \geq 2$, set $C = p^{a/e}q^{b/e}$, so $M = C^e$; this is precisely the same state space and witness set. Together with the four complete radicals and (25), the only conservative residual is

$$\mathcal{R} = \{(M,p,q,a,b) :\ \gcd(a,b) = 1,\ (a,b) \neq (1,1),\ pq \notin \{6,10,14,15\},\ M < C(p)\}.$$

$$C(p) = \begin{cases} 147456, & p = 2, \\ 262144, & p = 3, \\ 360000, & p \geq 5. \end{cases}$$

It contains all positive singleton and unequal valuations not paid by another stated rule. Unique factorization gives one ordered tuple per physical modulus. Since $M \geq p^2 q > p^3$, it is sufficient to enumerate primes $p \leq 71$, primes $q < C(p)/p^2$, and positive prime powers subject to the exact strict product test and these predicates. A second enumeration by factorization gives the same full list.

Inside this residual the ratio rule (26) has only the following possible supports:

(p, a, b)	prime q range
$(2, 2, 1)$	$4900 \leq q < 36864$
$(2, 3, 1)$	$2450 \leq q < 18432$
$(2, 4, 1)$	$3600 \leq q < 9216$
$(3, 2, 1)$	$11700 \leq q < 262144/9$
$(5, 2, 1)$	$10000 \leq q < 14400$

Indeed $q \geq c(p)p^a$ and $p^a q < C(p)$ imply $c(p)p^{2a} < C(p)$, while $a = 1, b = 1$ is excluded. The lower endpoints also impose $M \geq 19600$. This proves that the rule excludes no unseen exponent orientation.

The exact coverage table is

stage	rows	mathematical reason
conservative residual $\mathcal{R}$	19239	complete original predicate
ratio-paid part	7857	(26)
after the ratio filter	11382	$19239 - 7857$
analytically certified part	6972	negative upper from the five complete bounds
finite remainder	4410	$11382 - 6972$
both primes at least five	1087	every retained $Q_M < 0$
smaller prime three	1422	every retained $Q_M < 0$
smaller prime two	1901	every retained $Q_M < 0$
remaining two-prime obligations	0	$1087 + 1422 + 1901 = 4410$

The 6972 analytic rows are a conservative certificate partition, not a claim that all other exact analytic bounds or true means are nonnegative. Each row retains all five bound enclosures. For α, the rational tails

$$\sum_{j=1}^{128} j^{-2} + \frac{1}{129} < \zeta(2) < \sum_{j=1}^{128} j^{-2} + \frac{1}{128}$$

are inverted outward. Integer square-root floors give rational $x_L \leq \sqrt{M} \leq x_U$ with the same fixed k and $t_U < 1$. The gains use the lower bounds

$$\delta_L = \min_{z \in \{t_L, t_U\}} \frac{1 + 2z - 3z^2}{2} + \frac{t_L^3}{x_U}, \qquad G_L = \max(1,\, 2x_L - M + k^2).$$

The first uses concavity, the second the exact identity $G_x = 1 + (1 - t)(2k - 1 + t) \geq 1$. Their denominator is the exact M, never an approximate root squared. After lowering α, write

$$\Delta_L = C_L + s_L \ell_\star, \quad C_L = \alpha_L(5\rho/8 + 3/8) + \rho/2 - 1/2 - 3u^2/4, \quad s_L = \alpha_L \rho - 1/2 + 3u^2/4,$$

where $\ell_\star = \ell_L$ for $s_L \geq 0$ and ℓ_U otherwise. Positive error coefficients are enclosed upward, while their own nonnegative gains are enclosed downward. Strictly negative resulting uppers pay the analytic rows. No sign-selected domain or interpolation is used.

Every original two-prime modulus is thus covered by a preliminary theorem, an explicitly proved reduction, or one of the complete finite certificate sets. We conclude $\mu(p^a q^b) < 0$ for all

distinct primes and positive valuations. For $B = p^\gamma q^\delta$, $\gamma, \delta \geq 1$, $N \geq 1$, put $M = B^N$. The original height, state measure and full-unit marks are identical, and $\log M = N \log B$. Division by $\log B > 0$ therefore gives

$$\mathbb{E} \log_B R_{(B,N)} < N/2.$$

This conclusion is restricted to exactly two prime divisors. It neither transfers to bases with three prime factors nor asserts monotonicity, optimal margins, exact true means or novelty.

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